Hello Prospective Students of aBa's Math 395 course on the Beauty of Math,

This 66 page PDF contains the 1pg course outline and an excerpt from the text *Proofs from THE BOOK*. It is chapter one on the topic of number theory. This is one of 5 areas of math which we will explore in Part 1 of the course dealing with BEAUTIFUL PROOFS. You ABSOLUTELY need to have taken (even concurrently this Fall 2014) at least ONE of the following proof-writing courses: 314, 324, or 425.

By the way, there have been selective blank pages placed into the PDF to preserve the front and back pages as they appear in the book. So print this pdf DOUBLE-SIDED starting with pg.1 or pg.3 (if you want to start with the "cover" of the book – well, I made this cover, it's not the official cover of the book).

Part 2 of the course will be on combinatorics of permutations. There is a lot of beauty here too and this is part of my research area, so this part of the course will give you a flavor of what doing research with me might entail. Specifically in my research with a future student, I would like to extend the topics in part 2 to the world of *imprimitive complex reflection groups* (think of permutation matrices where the nonzero entries are r^{th} roots of unity). This group is a generalization of the symmetric group which you have seen in Math 312 and definitely in Math 425 (and in its permutation matrix form in Math 324). I will brush you up on this if it is foggy or missing completely from your memory.

Shalom,

aBa

Is Prof aBa "Power of brANDing" this course with me, the cat?

Prof aBa's Math 395 Directed Studies Course – Fall 2014

(1 credit course meeting 1 hour per week – time & place TBD)

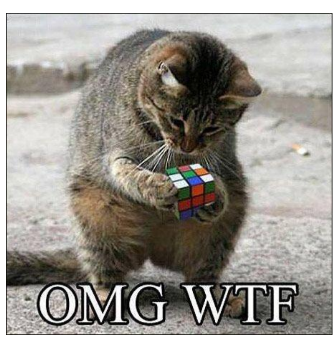
"Beautiful Proofs in Mathematics
and an Exploration of the Combinatorics of Permutations"

GOAL OF COURSE: There are two. First, we explore the most BEAUTIFUL proofs in mathematics. And I mean that! Second, we focus on combinatorial objects called permutations and study various combinatorial properties of these objects.

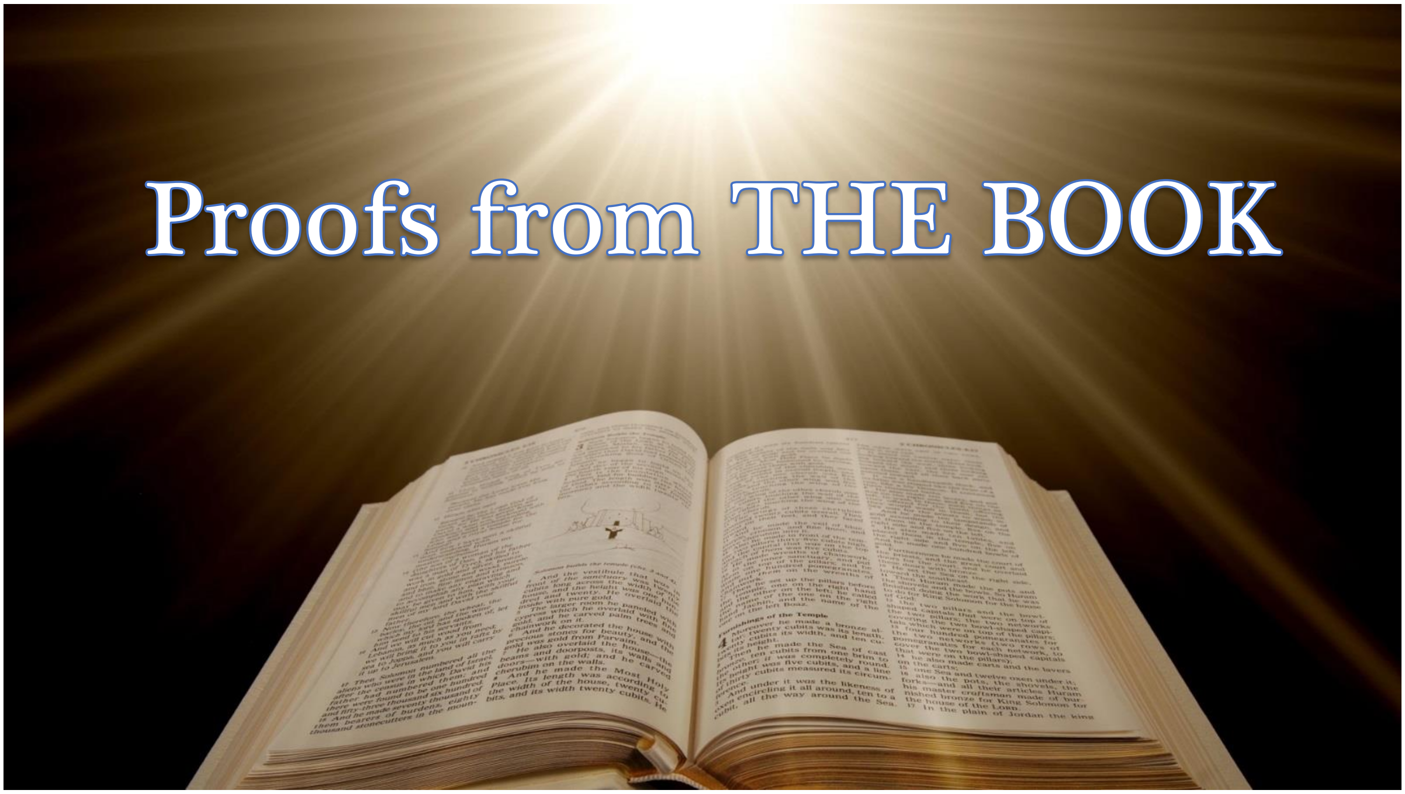
TEXTBOOKS: (I will provide excerpts of these for you as needed)
Proofs from THE BOOK (4th Edition) by Martin Aigner and Günther M. Ziegler (2010)
Combinatorics of Permutations by Miklos Bóna (2004)
The Symmetric Group (2nd Edition) by Bruce Sagan (2001)

FORMAT OF PART 1 OF THE COURSE:
Since this is a directed studies course, we will all take part in teaching it. The plan for the "Beauty of Mathematics" part of the course is for each of us to take turns presenting proofs from the book called *Proofs from THE BOOK*. **This part will be REALLY fun** because there are quite a variety of topics presented and EVERYTHING is beautiful. We will be learning some of the most fundamental results in areas of math from number theory, geometry, analysis, combinatorics, and graph theory.

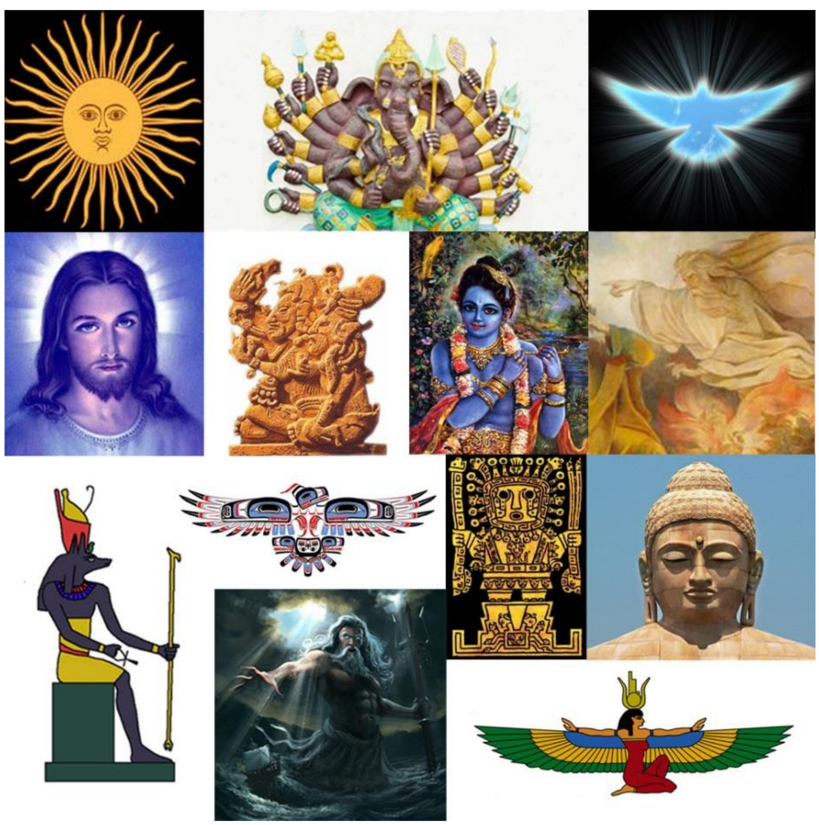
FORMAT OF PART 2 OF THE COURSE:
This will start out with me lecturing on permutations like in a standard lecture class. Then students present material. Time-permitting, we will explore possible directions of research that we can do next semester, next summer, next year, or whenever!



CAVEAT!!!: There being two parts to this course does NOT mean that Part 1 takes half the semester and Part 2 is the other half. If we are having LOADS of fun with Part 1, then we can just continue presenting the most beautiful proofs to one another. I can always teach Part 2 next semester, if that is the case! And if after the 1st two weeks or so, everyone is bored with beautiful proofs, then we can move on to Part 2.



Paul Erdős liked to talk about THE BOOK, in which God maintains the perfect proofs for mathematical theorems, following the dictum of G.H. Hardy that there is no permanent place for ugly mathematics.



"You do not have to believe in God but, as a mathematician, you should believe in the book." -Paul Erdős

Martin Aigner
Günter M. Ziegler

Proofs from THE BOOK

Fourth Edition

Including Illustrations by Karl H. Hofmann

Springer

Prof. Dr. Martin Aigner
FB Mathematik und Informatik
Freie Universität Berlin
Arnimallee 3
14195 Berlin
Deutschland
aigner@math.fu-berlin.de

Prof. Günter M. Ziegler
Institut für Mathematik, MA 6-2
Technische Universität Berlin
Straße des 17. Juni 136
10623 Berlin
Deutschland
ziegler@math.tu-berlin.de

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Preface

Paul Erdős liked to talk about The Book, in which God maintains the perfect proofs for mathematical theorems, following the dictum of G. H. Hardy that there is no permanent place for ugly mathematics. Erdős also said that you need not believe in God but, as a mathematician, you should believe in The Book. A few years ago, we suggested to him to write up a first (and very modest) approximation to The Book. He was enthusiastic about the idea and, characteristically, went to work immediately, filling page after page with his suggestions. Our book was supposed to appear in March 1998 as a present to Erdős' 85th birthday. With Paul's unfortunate death in the summer of 1996, he is not listed as a co-author. Instead this book is dedicated to his memory.

Paul Erdős

We have no definition or characterization of what constitutes a proof from The Book: all we offer here is the examples that we have selected, hoping that our readers will share our enthusiasm about brilliant ideas, clever insights and wonderful observations. We also hope that our readers will enjoy this despite the imperfections of our exposition. The selection is to a great extent influenced by Paul Erdős himself. A large number of the topics were suggested by him, and many of the proofs trace directly back to him, or were initiated by his supreme insight in asking the right question or in making the right conjecture. So to a large extent this book reflects the views of Paul Erdős as to what should be considered a proof from The Book.

"The Book"

A limiting factor for our selection of topics was that everything in this book is supposed to be accessible to readers whose backgrounds include only a modest amount of technique from undergraduate mathematics. A little linear algebra, some basic analysis and number theory, and a healthy dollop of elementary concepts and reasonings from discrete mathematics should be sufficient to understand and enjoy everything in this book.

We are extremely grateful to the many people who helped and supported us with this project — among them the students of a seminar where we discussed a preliminary version, to Benno Artmann, Stephan Brandt, Stefan Felsner, Eli Goodman, Torsten Heldmann, and Hans Mielke. We thank Margrit Barrett, Christian Bressler, Ewgenij Gawrilow, Michael Joswig, Elke Pose, and Jörg Rambau for their technical help in composing this book. We are in great debt to Tom Trotter who read the manuscript from first to last page, to Karl H. Hofmann for his wonderful drawings, and most of all to the late great Paul Erdős himself.

Berlin, March 1998 *Martin Aigner · Günter M. Ziegler*

Preface to the Fourth Edition

When we started this project almost fifteen years ago we could not possibly imagine what a wonderful and lasting response our book about The Book would have, with all the warm letters, interesting comments, new editions, and as of now thirteen translations. It is no exaggeration to say that it has become a part of our lives.

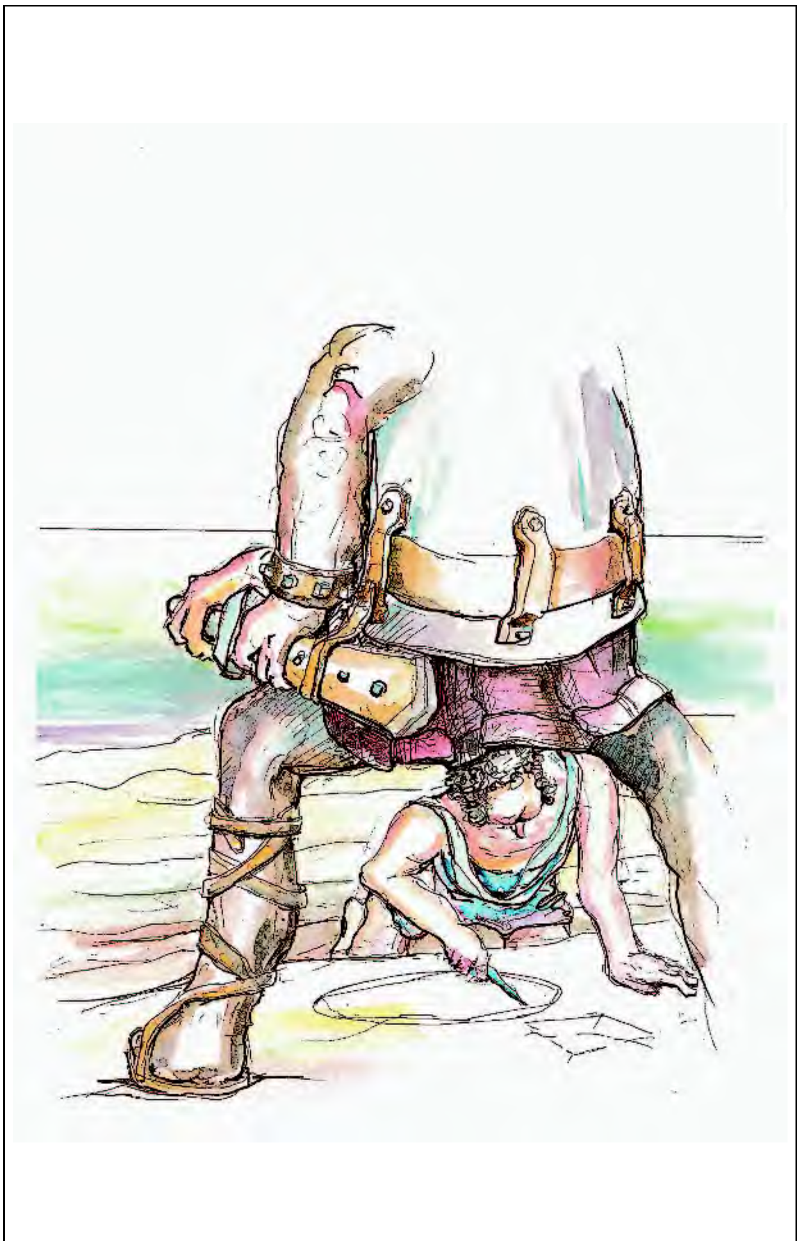
In addition to numerous improvements, partly suggested by our readers, the present fourth edition contains five new chapters: Two classics, the law of quadratic reciprocity and the fundamental theorem of algebra, two chapters on tiling problems and their intriguing solutions, and a highlight in graph theory, the chromatic number of Kneser graphs.

We thank everyone who helped and encouraged us over all these years: For the second edition this included Stephan Brandt, Christian Elsholtz, Jürgen Elstrodt, Daniel Grieser, Roger Heath-Brown, Lee L. Keener, Christian Lebœuf, Hanfried Lenz, Nicolas Puech, John Scholes, Bernulf Weißbach, and *many* others. The third edition benefitted especially from input by David Bevan, Anders Björner, Dietrich Braess, John Cosgrave, Hubert Kalf, Günter Pickert, Alistair Sinclair, and Herb Wilf. For the present edition, we are particularly grateful to contributions by France Dacar, Oliver Deiser, Anton Dochtermann, Michael Harbeck, Stefan Hougardy, Hendrik W. Lenstra, Günter Rote, Moritz Schmitt, and Carsten Schultz. Moreover, we thank Ruth Allewelt at Springer in Heidelberg as well as Christoph Eyrich, Torsten Heldmann, and Elke Pose in Berlin for their help and support throughout these years. And finally, this book would certainly not look the same without the original design suggested by Karl-Friedrich Koch, and the superb new drawings provided for each edition by Karl H. Hofmann.

Berlin, July 2009 *Martin Aigner · Günter M. Ziegler*

Table of Contents

Number Theory



"Irrationality and π"

Six proofs of the infinity of primes

Chapter 1

It is only natural that we start these notes with probably the oldest Book Proof, usually attributed to Euclid (*Elements* IX, 20). It shows that the sequence of primes does not end.

■ **Euclid's Proof.** For any finite set $\{p_1, \ldots, p_r\}$ of primes, consider the number $n = p_1 p_2 \cdots p_r + 1$. This n has a prime divisor p. But p is not one of the p_i: otherwise p would be a divisor of n and of the product $p_1 p_2 \cdots p_r$, and thus also of the difference $n - p_1 p_2 \cdots p_r = 1$, which is impossible. So a finite set $\{p_1, \ldots, p_r\}$ cannot be the collection of *all* prime numbers. $\square$

Before we continue let us fix some notation. $\mathbb{N} = \{1, 2, 3, \ldots\}$ is the set of natural numbers, $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$ the set of integers, and $\mathbb{P} = \{2, 3, 5, 7, \ldots\}$ the set of primes.

In the following, we will exhibit various other proofs (out of a much longer list) which we hope the reader will like as much as we do. Although they use different view-points, the following basic idea is common to all of them: The natural numbers grow beyond all bounds, and every natural number $n \geq 2$ has a prime divisor. These two facts taken together force $\mathbb{P}$ to be infinite. The next proof is due to Christian Goldbach (from a letter to Leonhard Euler 1730), the third proof is apparently folklore, the fourth one is by Euler himself, the fifth proof was proposed by Harry Fürstenberg, while the last proof is due to Paul Erdős.

■ **Second Proof.** Let us first look at the *Fermat numbers* $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, \ldots$. We will show that any two Fermat numbers are relatively prime; hence there must be infinitely many primes. To this end, we verify the recursion

$$\prod_{k=0}^{n-1} F_k \;=\; F_n - 2 \qquad (n \geq 1),$$

from which our assertion follows immediately. Indeed, if m is a divisor of, say, F_k and F_n $(k < n)$, then m divides 2, and hence $m = 1$ or 2. But $m = 2$ is impossible since all Fermat numbers are odd.

$$\begin{aligned} F_0 &= 3 \\ F_1 &= 5 \\ F_2 &= 17 \\ F_3 &= 257 \\ F_4 &= 65537 \\ F_5 &= 641 \cdot 6700417 \end{aligned}$$

The first few Fermat numbers

To prove the recursion we use induction on n. For $n = 1$ we have $F_0 = 3$ and $F_1 - 2 = 3$. With induction we now conclude

$$\prod_{k=0}^{n} F_k \;=\; \Big(\prod_{k=0}^{n-1} F_k\Big) F_n \;=\; (F_n - 2) F_n \;=$$
$$= (2^{2^n} - 1)(2^{2^n} + 1) \;=\; 2^{2^{n+1}} - 1 \;=\; F_{n+1} - 2. \qquad \square$$

Lagrange's Theorem

If G is a finite (multiplicative) group and U is a subgroup, then $|U|$ divides $|G|$.

■ **Proof.** Consider the binary relation

$$a \sim b :\iff ba^{-1} \in U.$$

It follows from the group axioms that $\sim$ is an equivalence relation. The equivalence class containing an element a is precisely the coset

$$Ua = \{xa : x \in U\}.$$

Since clearly $|Ua| = |U|$, we find that G decomposes into equivalence classes, all of size $|U|$, and hence that $|U|$ divides $|G|$. □

In the special case when U is a cyclic subgroup $\{a, a^2, \ldots, a^m\}$ we find that m (the smallest positive integer such that $a^m = 1$, called the *order* of a) divides the size $|G|$ of the group.

■ **Third Proof.** Suppose $\mathbb{P}$ is finite and p is the largest prime. We consider the so-called *Mersenne number* $2^p - 1$ and show that any prime factor q of $2^p - 1$ is bigger than p, which will yield the desired conclusion. Let q be a prime dividing $2^p - 1$, so we have $2^p \equiv 1 \pmod q$. Since p is prime, this means that the element 2 has order p in the multiplicative group $\mathbb{Z}_q \backslash \{0\}$ of the field $\mathbb{Z}_q$. This group has $q-1$ elements. By Lagrange's theorem (see the box) we know that the order of every element divides the size of the group, that is, we have $p \mid q-1$, and hence $p < q$. □

Now let us look at a proof that uses elementary calculus.

■ **Fourth Proof.** Let $\pi(x) := \#\{p \le x : p \in \mathbb{P}\}$ be the number of primes that are less than or equal to the real number x. We number the primes $\mathbb{P} = \{p_1, p_2, p_3, \ldots\}$ in increasing order. Consider the natural logarithm $\log x$, defined as $\log x = \int_1^x \frac{1}{t} dt$.

Now we compare the area below the graph of $f(t) = \frac{1}{t}$ with an upper step function. (See also the appendix on page 10 for this method.) Thus for $n \le x < n+1$ we have

$$\begin{aligned} \log x \quad &\le \quad 1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{n-1} + \frac{1}{n} \\ &\le \quad \sum \frac{1}{m}, \text{ where the sum extends over all } m \in \mathbb{N} \text{ which have only prime divisors } p \le x. \end{aligned}$$

Since every such m can be written in a *unique* way as a product of the form $\prod\limits_{p \le x} p^{k_p}$, we see that the last sum is equal to

$$\prod_{\substack{p \in \mathbb{P} \\ p \le x}} \Big(\sum_{k \ge 0} \frac{1}{p^k} \Big).$$

The inner sum is a geometric series with ratio $\frac{1}{p}$, hence

$$\log x \;\le\; \prod_{\substack{p \in \mathbb{P} \\ p \le x}} \frac{1}{1 - \frac{1}{p}} \;=\; \prod_{\substack{p \in \mathbb{P} \\ p \le x}} \frac{p}{p-1} \;=\; \prod_{k=1}^{\pi(x)} \frac{p_k}{p_k - 1}.$$

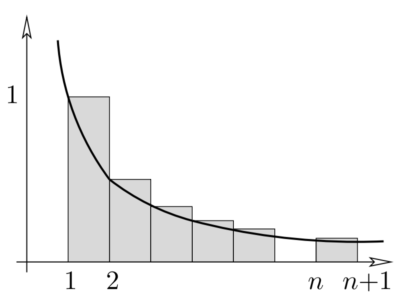
Now clearly $p_k \ge k+1$, and thus

$$\frac{p_k}{p_k - 1} \;=\; 1 + \frac{1}{p_k - 1} \;\le\; 1 + \frac{1}{k} \;=\; \frac{k+1}{k},$$

and therefore

$$\log x \;\le\; \prod_{k=1}^{\pi(x)} \frac{k+1}{k} \;=\; \pi(x) + 1.$$

Everybody knows that $\log x$ is not bounded, so we conclude that $\pi(x)$ is unbounded as well, and so there are infinitely many primes. □



Steps above the function $f(t) = \frac{1}{t}$

■ **Fifth Proof.** After analysis it's topology now! Consider the following curious topology on the set $\mathbb{Z}$ of integers. For $a, b \in \mathbb{Z}$, $b > 0$, we set

$$N_{a,b} = \{a + nb : n \in \mathbb{Z}\}.$$

Each set $N_{a,b}$ is a two-way infinite arithmetic progression. Now call a set $O \subseteq \mathbb{Z}$ *open* if either O is empty, or if to every $a \in O$ there exists some $b > 0$ with $N_{a,b} \subseteq O$. Clearly, the union of open sets is open again. If O_1, O_2 are open, and $a \in O_1 \cap O_2$ with $N_{a,b_1} \subseteq O_1$ and $N_{a,b_2} \subseteq O_2$, then $a \in N_{a,b_1 b_2} \subseteq O_1 \cap O_2$. So we conclude that any finite intersection of open sets is again open. So, this family of open sets induces a bona fide topology on $\mathbb{Z}$.

Let us note two facts:

(A) Any nonempty open set is infinite.

(B) Any set $N_{a,b}$ is closed as well.

Indeed, the first fact follows from the definition. For the second we observe

$$N_{a,b} \;=\; \mathbb{Z} \setminus \bigcup_{i=1}^{b-1} N_{a+i,b},$$

which proves that $N_{a,b}$ is the complement of an open set and hence closed.

"Pitching flat rocks, infinitely"

So far the primes have not yet entered the picture — but here they come. Since any number $n \neq 1, -1$ has a prime divisor p, and hence is contained in $N_{0,p}$, we conclude

$$\mathbb{Z} \setminus \{1, -1\} \;=\; \bigcup_{p \in \mathbb{P}} N_{0,p}.$$

Now if $\mathbb{P}$ were finite, then $\bigcup_{p\in\mathbb{P}} N_{0,p}$ would be a finite union of closed sets (by (B)), and hence closed. Consequently, $\{1, -1\}$ would be an open set, in violation of (A). □

■ **Sixth Proof.** Our final proof goes a considerable step further and demonstrates not only that there are infinitely many primes, but also that the series $\sum_{p\in\mathbb{P}} \frac{1}{p}$ diverges. The first proof of this important result was given by Euler (and is interesting in its own right), but our proof, devised by Erdős, is of compelling beauty.

Let $p_1, p_2, p_3, \ldots$ be the sequence of primes in increasing order, and assume that $\sum_{p\in\mathbb{P}} \frac{1}{p}$ converges. Then there must be a natural number k such that $\sum_{i\geq k+1} \frac{1}{p_i} < \frac{1}{2}$. Let us call $p_1, \ldots, p_k$ the *small* primes, and $p_{k+1}, p_{k+2}, \ldots$ the *big* primes. For an arbitrary natural number N we therefore find

$$\sum_{i\geq k+1} \frac{N}{p_i} \;<\; \frac{N}{2}. \qquad (1)$$

Let N_b be the number of positive integers $n \leq N$ which are divisible by at least one big prime, and N_s the number of positive integers $n \leq N$ which have only small prime divisors. We are going to show that for a suitable N

$$N_b + N_s \;<\; N,$$

which will be our desired contradiction, since by definition $N_b + N_s$ would have to be equal to N.

To estimate N_b note that $\lfloor \frac{N}{p_i} \rfloor$ counts the positive integers $n \leq N$ which are multiples of p_i. Hence by (1) we obtain

$$N_b \;\leq\; \sum_{i \geq k+1} \left\lfloor \frac{N}{p_i} \right\rfloor \;<\; \frac{N}{2}. \tag{2}$$

Let us now look at N_s. We write every $n \leq N$ which has only small prime divisors in the form $n = a_n b_n^2$, where a_n is the square-free part. Every a_n is thus a product of *different* small primes, and we conclude that there are precisely 2^k different square-free parts. Furthermore, as $b_n \leq \sqrt{n} \leq \sqrt{N}$, we find that there are at most $\sqrt{N}$ different square parts, and so

$$N_s \;\leq\; 2^k \sqrt{N}.$$

Since (2) holds for *any* N, it remains to find a number N with $2^k \sqrt{N} \leq \frac{N}{2}$ or $2^{k+1} \leq \sqrt{N}$, and for this $N = 2^{2k+2}$ will do. $\square$

References

[1] B. ARTMANN: *Euclid — The Creation of Mathematics,* Springer-Verlag, New York 1999.

[2] P. ERDŐS: *Über die Reihe* $\sum \frac{1}{p}$, Mathematica, Zutphen B **7** (1938), 1-2.

[3] L. EULER: *Introductio in Analysin Infinitorum,* Tomus Primus, Lausanne 1748; Opera Omnia, Ser. 1, Vol. 8.

[4] H. FÜRSTENBERG: *On the infinitude of primes,* Amer. Math. Monthly **62** (1955), 353.

Bertrand's postulate

Chapter 2

We have seen that the sequence of prime numbers $2, 3, 5, 7, \ldots$ is infinite. To see that the size of its gaps is not bounded, let $N := 2 \cdot 3 \cdot 5 \cdots p$ denote the product of all prime numbers that are smaller than $k+2$, and note that none of the k numbers

$$N+2, N+3, N+4, \ldots, N+k, N+(k+1)$$

is prime, since for $2 \leq i \leq k+1$ we know that i has a prime factor that is smaller than $k+2$, and this factor also divides N, and hence also $N+i$. With this recipe, we find, for example, for $k = 10$ that none of the ten numbers

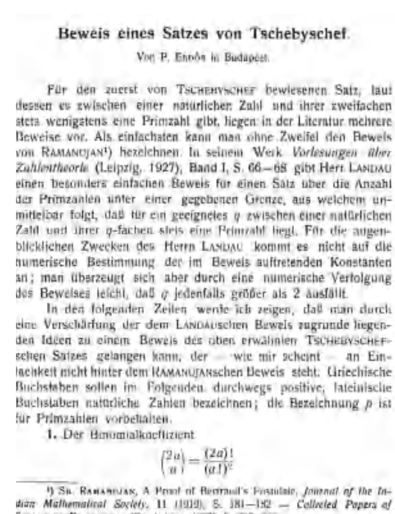
$$2312, 2313, 2314, \ldots, 2321$$

is prime.

Joseph Bertrand

But there are also upper bounds for the gaps in the sequence of prime numbers. A famous bound states that "the gap to the next prime cannot be larger than the number we start our search at." This is known as Bertrand's postulate, since it was conjectured and verified empirically for $n < 3\,000\,000$ by Joseph Bertrand. It was first proved for all n by Pafnuty Chebyshev in 1850. A much simpler proof was given by the Indian genius Ramanujan. Our Book Proof is by Paul Erdős: it is taken from Erdős' first published paper, which appeared in 1932, when Erdős was 19.

Bertrand's postulate.
For every $n \geq 1$, there is some prime number p with $n < p \leq 2n$.



Beweis eines Satzes von Tschebyschef.

Von P. Erdős in Budapest.

Für den zuerst von TSCHEBYSCHEF bewiesenen Satz, laut dessen es zwischen einer natürlichen Zahl und ihrer zweifachen stets wenigstens eine Primzahl gibt, liegen in der Literatur mehrere Beweise vor. Als einfachsten kann man ohne Zweifel den Beweis von RAMANUJAN[1]) bezeichnen. In seinem Werk *Vorlesungen über Zahlentheorie* (Leipzig, 1927), Band I, S. 66—68 gibt Herr LANDAU einen besonders einfachen Beweis für einen Satz über die Anzahl der Primzahlen unter einer gegebenen Grenze, aus welchem unmittelbar folgt, daß für ein geeignetes q zwischen einer natürlichen Zahl und ihrer q-fachen stets eine Primzahl liegt. Für die augenblicklichen Zwecken des Herrn LANDAU kommt es nicht auf die numerische Bestimmung der im Beweis auftretenden Konstanten an; man überzeugt sich aber durch eine numerische Verfolgung des Beweises leicht, daß q jedenfalls größer als 2 ausfällt.

In den folgenden Zeilen werde ich zeigen, daß man durch eine Verschärfung der dem LANDAUschen Beweis zugrunde liegenden Ideen zu einem Beweis des oben erwähnten TSCHEBYSCHEFschen Satzes gelangen kann, der — wie mir scheint — an Einfachheit nicht hinter dem RAMANUJANschen Beweis steht. Griechische Buchstaben sollen im Folgenden durchwegs positive, lateinische Buchstaben natürliche Zahlen bezeichnen; die Bezeichnung p ist für Primzahlen vorbehalten.

1. Der Binomialkoeffizient

$$\binom{2a}{a} = \frac{(2a)!}{(a!)^2}$$

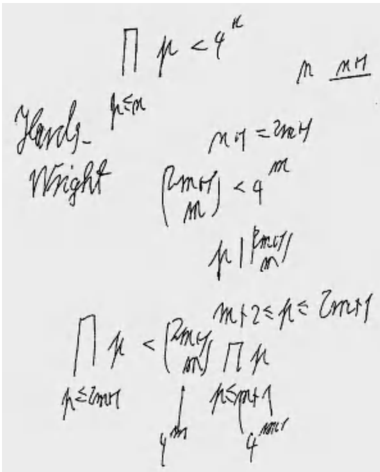
[1]) S. RAMANUJAN, A Proof of Bertrand's Postulate, *Journal of the Indian Mathematical Society*, 11 (1919), S. 181—182 — *Collected Papers of Srinivasa Ramanujan* (Cambridge, 1927), S. 208—209.

■ **Proof.** We will estimate the size of the binomial coefficient $\binom{2n}{n}$ carefully enough to see that if it didn't have any prime factors in the range $n < p \leq 2n$, then it would be "too small." Our argument is in five steps.

(1) We first prove Bertrand's postulate for $n < 4000$. For this one does not need to check 4000 cases: it suffices (this is "Landau's trick") to check that

$$2, 3, 5, 7, 13, 23, 43, 83, 163, 317, 631, 1259, 2503, 4001$$

is a sequence of prime numbers, where each is smaller than twice the previous one. Hence every interval $\{y : n < y \leq 2n\}$, with $n \leq 4000$, contains one of these 14 primes.



(2) Next we prove that

$$\prod_{p \le x} p \;\le\; 4^{x-1} \qquad \text{for all real } x \ge 2, \tag{1}$$

where our notation — here and in the following — is meant to imply that the product is taken over all *prime* numbers $p \le x$. The proof that we present for this fact uses induction on the number of these primes. It is not from Erdős' original paper, but it is also due to Erdős (see the margin), and it is a true Book Proof. First we note that if q is the largest prime with $q \le x$, then

$$\prod_{p \le x} p \;=\; \prod_{p \le q} p \qquad \text{and} \qquad 4^{q-1} \;\le\; 4^{x-1}.$$

Thus it suffices to check (1) for the case where $x = q$ is a prime number. For $q = 2$ we get "$2 \le 4$," so we proceed to consider odd primes $q = 2m+1$. (Here we may assume, by induction, that (1) is valid for all integers x in the set $\{2, 3, \ldots, 2m\}$.) For $q = 2m+1$ we split the product and compute

$$\prod_{p \le 2m+1} p \;=\; \prod_{p \le m+1} p \cdot \prod_{m+1 < p \le 2m+1} p \;\le\; 4^m \binom{2m+1}{m} \;\le\; 4^m 2^{2m} \;=\; 4^{2m}.$$

All the pieces of this "one-line computation" are easy to see. In fact,

$$\prod_{p \le m+1} p \;\le\; 4^m$$

holds by induction. The inequality

$$\prod_{m+1 < p \le 2m+1} p \;\le\; \binom{2m+1}{m}$$

follows from the observation that $\binom{2m+1}{m} = \frac{(2m+1)!}{m!(m+1)!}$ is an integer, where the primes that we consider all are factors of the numerator $(2m+1)!$, but not of the denominator $m!(m+1)!$. Finally

$$\binom{2m+1}{m} \;\le\; 2^{2m}$$

holds since

$$\binom{2m+1}{m} \quad \text{and} \quad \binom{2m+1}{m+1}$$

are two (equal!) summands that appear in

$$\sum_{k=0}^{2m+1} \binom{2m+1}{k} = 2^{2m+1}.$$

(3) From Legendre's theorem (see the box) we get that $\binom{2n}{n} = \frac{(2n)!}{n!n!}$ contains the prime factor p exactly

$$\sum_{k \ge 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2 \left\lfloor \frac{n}{p^k} \right\rfloor \right)$$

Legendre's theorem

The number $n!$ contains the prime factor p exactly

$$\sum_{k \ge 1} \left\lfloor \frac{n}{p^k} \right\rfloor$$

times.

■ **Proof.** Exactly $\lfloor \frac{n}{p} \rfloor$ of the factors of $n! = 1 \cdot 2 \cdot 3 \cdots n$ are divisible by p, which accounts for $\lfloor \frac{n}{p} \rfloor$ p-factors. Next, $\lfloor \frac{n}{p^2} \rfloor$ of the factors of $n!$ are even divisible by p^2, which accounts for the next $\lfloor \frac{n}{p^2} \rfloor$ prime factors p of $n!$, etc. □

times. Here each summand is at most 1, since it satisfies

$$\left\lfloor \frac{2n}{p^k} \right\rfloor - 2\left\lfloor \frac{n}{p^k} \right\rfloor \;<\; \frac{2n}{p^k} - 2\left(\frac{n}{p^k} - 1\right) \;=\; 2,$$

and it is an integer. Furthermore the summands vanish whenever $p^k > 2n$. Thus $\binom{2n}{n}$ contains p exactly

$$\sum_{k \geq 1} \left(\left\lfloor \frac{2n}{p^k} \right\rfloor - 2\left\lfloor \frac{n}{p^k} \right\rfloor \right) \;\leq\; \max\{r : p^r \leq 2n\}$$

times. Hence the largest power of p that divides $\binom{2n}{n}$ is not larger than $2n$. In particular, primes $p > \sqrt{2n}$ appear at most once in $\binom{2n}{n}$.

Furthermore — and this, according to Erdős, is the key fact for his proof — primes p that satisfy $\frac{2}{3}n < p \leq n$ do not divide $\binom{2n}{n}$ at all! Indeed, $3p > 2n$ implies (for $n \geq 3$, and hence $p \geq 3$) that p and $2p$ are the only multiples of p that appear as factors in the numerator of $\frac{(2n)!}{n!n!}$, while we get two p-factors in the denominator.

Examples such as
$\binom{26}{13} = 2^3 \cdot 5^2 \cdot 7 \cdot 17 \cdot 19 \cdot 23$
$\binom{28}{14} = 2^3 \cdot 3^3 \cdot 5^2 \cdot 17 \cdot 19 \cdot 23$
$\binom{30}{15} = 2^4 \cdot 3^2 \cdot 5 \cdot 17 \cdot 19 \cdot 23 \cdot 29$
illustrate that "very small" prime factors $p < \sqrt{2n}$ can appear as higher powers in $\binom{2n}{n}$, "small" primes with $\sqrt{2n} < p \leq \frac{2}{3}n$ appear at most once, while factors in the gap with $\frac{2}{3}n < p \leq n$ don't appear at all.

(4) Now we are ready to estimate $\binom{2n}{n}$. For $n \geq 3$, using an estimate from page 12 for the lower bound, we get

$$\frac{4^n}{2n} \;\leq\; \binom{2n}{n} \;\leq\; \prod_{p \leq \sqrt{2n}} 2n \;\cdot \prod_{\sqrt{2n} < p \leq \frac{2}{3}n} p \;\cdot \prod_{n < p \leq 2n} p$$

and thus, since there are not more than $\sqrt{2n}$ primes $p \leq \sqrt{2n}$,

$$4^n \;\leq\; (2n)^{1+\sqrt{2n}} \;\cdot \prod_{\sqrt{2n} < p \leq \frac{2}{3}n} p \;\cdot \prod_{n < p \leq 2n} p \qquad \text{for} \quad n \geq 3. \qquad (2)$$

(5) Assume now that there is no prime p with $n < p \leq 2n$, so the second product in (2) is 1. Substituting (1) into (2) we get

$$4^n \;\leq\; (2n)^{1+\sqrt{2n}}\, 4^{\frac{2}{3}n}$$

or

$$4^{\frac{1}{3}n} \;\leq\; (2n)^{1+\sqrt{2n}}, \qquad (3)$$

which is false for n large enough! In fact, using $a + 1 < 2^a$ (which holds for all $a \geq 2$, by induction) we get

$$2n = \left(\sqrt[6]{2n}\right)^6 < \left(\left\lfloor \sqrt[6]{2n} \right\rfloor + 1\right)^6 < 2^{6\left\lfloor \sqrt[6]{2n} \right\rfloor} \leq 2^{6\sqrt[6]{2n}}, \qquad (4)$$

and thus for $n \geq 50$ (and hence $18 < 2\sqrt{2n}$) we obtain from (3) and (4)

$$2^{2n} \leq (2n)^{3\left(1+\sqrt{2n}\right)} < 2^{\sqrt[6]{2n}\left(18+18\sqrt{2n}\right)} < 2^{20\sqrt[6]{2n}\sqrt{2n}} = 2^{20(2n)^{2/3}}.$$

This implies $(2n)^{1/3} < 20$, and thus $n < 4000$. $\square$

One can extract even more from this type of estimates: From (2) one can derive with the same methods that

$$\prod_{n<p\leq 2n} p \geq 2^{\frac{1}{30}n} \quad \text{for} \quad n \geq 4000,$$

and thus that there are at least

$$\log_{2n}\left(2^{\frac{1}{30}n}\right) = \frac{1}{30}\frac{n}{\log_2 n + 1}$$

primes in the range between n and $2n$.

This is not that bad an estimate: the "true" number of primes in this range is roughly $n/\log n$. This follows from the "prime number theorem," which says that the limit

$$\lim_{n\to\infty} \frac{\#\{p \leq n : p \text{ is prime}\}}{n/\log n}$$

exists, and equals 1. This famous result was first proved by Hadamard and de la Vallée-Poussin in 1896; Selberg and Erdős found an elementary proof (without complex analysis tools, but still long and involved) in 1948.

On the prime number theorem itself the final word, it seems, is still not in: for example a proof of the Riemann hypothesis (see page 49), one of the major unsolved open problems in mathematics, would also give a substantial improvement for the estimates of the prime number theorem. But also for Bertrand's postulate, one could expect dramatic improvements. In fact, the following is a famous unsolved problem:

Is there always a prime between n^2 and $(n+1)^2$?

For additional information see [3, p. 19] and [4, pp. 248, 257].

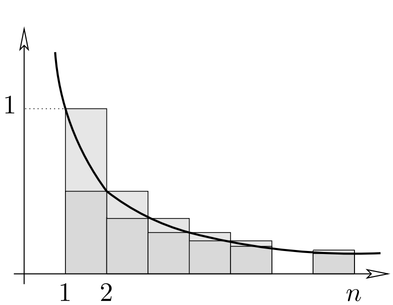
Appendix: Some estimates

Estimating via integrals

There is a very simple-but-effective method of estimating sums by integrals (as already encountered on page 4). For estimating the *harmonic numbers*

$$H_n = \sum_{k=1}^{n} \frac{1}{k}$$

we draw the figure in the margin and derive from it



$$H_n - 1 = \sum_{k=2}^{n} \frac{1}{k} < \int_1^n \frac{1}{t}\,dt = \log n$$

by comparing the area below the graph of $f(t) = \frac{1}{t}$ $(1 \leq t \leq n)$ with the area of the dark shaded rectangles, and

$$H_n - \frac{1}{n} = \sum_{k=1}^{n-1} \frac{1}{k} > \int_1^n \frac{1}{t}\,dt = \log n$$

by comparing with the area of the large rectangles (including the lightly shaded parts). Taken together, this yields

$$\log n + \frac{1}{n} \;<\; H_n \;<\; \log n \;+\; 1.$$

In particular, $\lim_{n\to\infty} H_n \to \infty$, and the order of growth of H_n is given by $\lim_{n\to\infty} \frac{H_n}{\log n} = 1$. But much better estimates are known (see [2]), such as

$$H_n \;=\; \log n + \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \frac{1}{120n^4} + O\left(\frac{1}{n^6}\right),$$

Here $O\left(\frac{1}{n^6}\right)$ denotes a function $f(n)$ such that $f(n) \leq c\frac{1}{n^6}$ holds for some constant c.

where $\gamma \approx 0.5772$ is "Euler's constant."

Estimating factorials — Stirling's formula

The same method applied to

$$\log(n!) \;=\; \log 2 + \log 3 + \ldots + \log n \;=\; \sum_{k=2}^{n} \log k$$

yields

$$\log((n-1)!) \;<\; \int_1^n \log t\, dt \;<\; \log(n!),$$

where the integral is easily computed:

$$\int_1^n \log t\, dt \;=\; \Big[t\log t - t\Big]_1^n \;=\; n\log n - n + 1.$$

Thus we get a lower estimate on $n!$

$$n! \;>\; e^{n\log n - n + 1} \;=\; e\left(\frac{n}{e}\right)^n$$

and at the same time an upper estimate

$$n! \;=\; n\,(n-1)! \;<\; ne^{n\log n - n + 1} \;=\; en\left(\frac{n}{e}\right)^n.$$

Here a more careful analysis is needed to get the asymptotics of $n!$, as given by *Stirling's formula*

$$n! \;\sim\; \sqrt{2\pi n}\left(\frac{n}{e}\right)^n.$$

Here $f(n) \sim g(n)$ means that $\lim_{n\to\infty} \frac{f(n)}{g(n)} = 1.$

And again there are more precise versions available, such as

$$n! \;=\; \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \left(1 + \frac{1}{12n} + \frac{1}{288n^2} - \frac{139}{5140n^3} + O\left(\frac{1}{n^4}\right)\right).$$

Estimating binomial coefficients

Just from the definition of the binomial coefficients $\binom{n}{k}$ as the number of k-subsets of an n-set, we know that the sequence $\binom{n}{0}, \binom{n}{1}, \ldots, \binom{n}{n}$ of binomial coefficients

- sums to $\sum\limits_{k=0}^{n} \binom{n}{k} = 2^n$
- is symmetric: $\binom{n}{k} = \binom{n}{n-k}$.

$$\begin{array}{ccccccccccccccc} &&&&&&&1&&&&&&& \\ &&&&&&1&&1&&&&&& \\ &&&&&1&&2&&1&&&&& \\ &&&&1&&3&&3&&1&&&& \\ &&&1&&4&&6&&4&&1&&& \\ &&1&&5&&10&&10&&5&&1&& \\ &1&&6&&15&&20&&15&&6&&1& \\ 1&&7&&21&&35&&35&&21&&7&&1 \end{array}$$

Pascal's triangle

From the functional equation $\binom{n}{k} = \frac{n-k+1}{k}\binom{n}{k-1}$ one easily finds that for every n the binomial coefficients $\binom{n}{k}$ form a sequence that is symmetric and *unimodal*: it increases towards the middle, so that the middle binomial coefficients are the largest ones in the sequence:

$$1 = \binom{n}{0} < \binom{n}{1} < \cdots < \binom{n}{\lfloor n/2 \rfloor} = \binom{n}{\lceil n/2 \rceil} > \cdots > \binom{n}{n-1} > \binom{n}{n} = 1.$$

Here $\lfloor x \rfloor$ resp. $\lceil x \rceil$ denotes the number x rounded down resp. rounded up to the nearest integer.

From the asymptotic formulas for the factorials mentioned above one can obtain very precise estimates for the sizes of binomial coefficients. However, we will only need very weak and simple estimates in this book, such as the following: $\binom{n}{k} \le 2^n$ for all k, while for $n \ge 2$ we have

$$\binom{n}{\lfloor n/2 \rfloor} \;\ge\; \frac{2^n}{n},$$

with equality only for $n = 2$. In particular, for $n \ge 1$,

$$\binom{2n}{n} \;\ge\; \frac{4^n}{2n}.$$

This holds since $\binom{n}{\lfloor n/2 \rfloor}$, a middle binomial coefficient, is the largest entry in the sequence $\binom{n}{0} + \binom{n}{n}, \binom{n}{1}, \binom{n}{2}, \ldots, \binom{n}{n-1}$, whose sum is 2^n, and whose average is thus $\frac{2^n}{n}$.

On the other hand, we note the upper bound for binomial coefficients

$$\binom{n}{k} \;=\; \frac{n(n-1)\cdots(n-k+1)}{k!} \;\le\; \frac{n^k}{k!} \;\le\; \frac{n^k}{2^{k-1}},$$

which is a reasonably good estimate for the "small" binomial coefficients at the tails of the sequence, when n is large (compared to k).

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Binomial coefficients are (almost) never powers

Chapter 3

There is an epilogue to Bertrand's postulate which leads to a beautiful result on binomial coefficients. In 1892 Sylvester strengthened Bertrand's postulate in the following way:

> *If $n \geq 2k$, then at least one of the numbers $n, n-1, \ldots, n-k+1$ has a prime divisor p greater than k.*

Note that for $n = 2k$ we obtain precisely Bertrand's postulate. In 1934, Erdős gave a short and elementary Book Proof of Sylvester's result, running along the lines of his proof of Bertrand's postulate. There is an equivalent way of stating Sylvester's theorem:

> *The binomial coefficient*
>
> $$\binom{n}{k} = \frac{n(n-1)\cdots(n-k+1)}{k!} \qquad (n \geq 2k)$$
>
> *always has a prime factor $p > k$.*

With this observation in mind, we turn to another one of Erdős' jewels. When is $\binom{n}{k}$ equal to a power m^ℓ? It is easy to see that there are infinitely many solutions for $k = \ell = 2$, that is, of the equation $\binom{n}{2} = m^2$. Indeed, if $\binom{n}{2}$ is a square, then so is $\binom{(2n-1)^2}{2}$. To see this, set $n(n-1) = 2m^2$. It follows that

$$(2n-1)^2((2n-1)^2-1) = (2n-1)^2 4n(n-1) = 2(2m(2n-1))^2,$$

and hence

$$\binom{(2n-1)^2}{2} = (2m(2n-1))^2.$$

Beginning with $\binom{9}{2} = 6^2$ we thus obtain infinitely many solutions — the next one is $\binom{289}{2} = 204^2$. However, this does not yield all solutions. For example, $\binom{50}{2} = 35^2$ starts another series, as does $\binom{1682}{2} = 1189^2$. For $k = 3$ it is known that $\binom{n}{3} = m^2$ has the unique solution $n = 50, m = 140$. But now we are at the end of the line. For $k \geq 4$ and any $\ell \geq 2$ no solutions exist, and this is what Erdős proved by an ingenious argument.

$\binom{50}{3} = 140^2$
is the only solution for $k = 3$, $\ell = 2$

Theorem. *The equation $\binom{n}{k} = m^\ell$ has no integer solutions with $\ell \geq 2$ and $4 \leq k \leq n-4$.*

■ **Proof.** Note first that we may assume $n \geq 2k$ because of $\binom{n}{k} = \binom{n}{n-k}$. Suppose the theorem is false, and that $\binom{n}{k} = m^\ell$. The proof, by contradiction, proceeds in the following four steps.

(1) By Sylvester's theorem, there is a prime factor p of $\binom{n}{k}$ greater than k, hence p^ℓ divides $n(n-1)\cdots(n-k+1)$. Clearly, only one of the factors $n-i$ can be a multiple of p (because of $p > k$), and we conclude $p^\ell \mid n-i$, and therefore

$$n \;\geq\; p^\ell \;>\; k^\ell \;\geq\; k^2.$$

(2) Consider any factor $n-j$ of the numerator and write it in the form $n-j = a_j m_j^\ell$, where a_j is not divisible by any nontrivial ℓ-th power. We note by **(1)** that a_j has only prime divisors less than or equal to k. We want to show next that $a_i \neq a_j$ for $i \neq j$. Assume to the contrary that $a_i = a_j$ for some $i < j$. Then $m_i \geq m_j + 1$ and

$$\begin{aligned} k &> (n-i)-(n-j) \;=\; a_j(m_i^\ell - m_j^\ell) \;\geq\; a_j((m_j+1)^\ell - m_j^\ell) \\ &> a_j \ell m_j^{\ell-1} \;\geq\; \ell(a_j m_j^\ell)^{1/2} \;\geq\; \ell(n-k+1)^{1/2} \\ &\geq \ell(\tfrac{n}{2}+1)^{1/2} \;>\; n^{1/2}, \end{aligned}$$

which contradicts $n > k^2$ from above.

(3) Next we prove that the a_i's are the integers $1, 2, \ldots, k$ in some order. (According to Erdős, this is the crux of the proof.) Since we already know that they are all distinct, it suffices to prove that

$$a_0 a_1 \cdots a_{k-1} \quad \text{divides} \quad k!.$$

Substituting $n-j = a_j m_j^\ell$ into the equation $\binom{n}{k} = m^\ell$, we obtain

$$a_0 a_1 \cdots a_{k-1}(m_0 m_1 \cdots m_{k-1})^\ell \;=\; k! m^\ell.$$

Cancelling the common factors of $m_0 \cdots m_{k-1}$ and m yields

$$a_0 a_1 \cdots a_{k-1} u^\ell \;=\; k! v^\ell$$

with $\gcd(u,v) = 1$. It remains to show that $v = 1$. If not, then v contains a prime divisor p. Since $\gcd(u,v) = 1$, p must be a prime divisor of $a_0 a_1 \cdots a_{k-1}$ and hence is less than or equal to k. By the theorem of Legendre (see page 8) we know that $k!$ contains p to the power $\sum_{i\geq 1} \lfloor \frac{k}{p^i} \rfloor$. We now estimate the exponent of p in $n(n-1)\cdots(n-k+1)$. Let i be a positive integer, and let $b_1 < b_2 < \cdots < b_s$ be the multiples of p^i among $n, n-1, \ldots, n-k+1$. Then $b_s = b_1 + (s-1)p^i$ and hence

$$(s-1)p^i \;=\; b_s - b_1 \;\leq\; n-(n-k+1) \;=\; k-1,$$

which implies

$$s \;\leq\; \left\lfloor \frac{k-1}{p^i} \right\rfloor + 1 \;\leq\; \left\lfloor \frac{k}{p^i} \right\rfloor + 1.$$

So for each i the number of multiples of p^i among $n, \dots, n-k+1$, and hence among the a_j's, is bounded by $\lfloor \frac{k}{p^i} \rfloor + 1$. This implies that the exponent of p in $a_0 a_1 \cdots a_{k-1}$ is at most

$$\sum_{i=1}^{\ell-1} \Big(\Big\lfloor \frac{k}{p^i} \Big\rfloor + 1 \Big)$$

with the reasoning that we used for Legendre's theorem in Chapter 2. The only difference is that this time the sum stops at $i = \ell - 1$, since the a_j's contain no ℓ-th powers.

Taking both counts together, we find that the exponent of p in v^ℓ is at most

$$\sum_{i=1}^{\ell-1} \Big(\Big\lfloor \frac{k}{p^i} \Big\rfloor + 1 \Big) - \sum_{i \geq 1} \Big\lfloor \frac{k}{p^i} \Big\rfloor \;\; \leq \;\; \ell - 1,$$

and we have our desired contradiction, since v^ℓ is an ℓ-th power.

This suffices already to settle the case $\ell = 2$. Indeed, since $k \geq 4$ one of the a_i's must be equal to 4, but the a_i's contain no squares. So let us now assume that $\ell \geq 3$.

We see that our analysis so far agrees with $\binom{50}{3} = 140^2$, as

$$\begin{aligned} 50 &= 2 \cdot 5^2 \\ 49 &= 1 \cdot 7^2 \\ 48 &= 3 \cdot 4^2 \end{aligned}$$

and $5 \cdot 7 \cdot 4 = 140$.

(4) Since $k \geq 4$, we must have $a_{i_1} = 1, a_{i_2} = 2, a_{i_3} = 4$ for some i_1, i_2, i_3, that is,

$$n - i_1 = m_1^\ell, \;\; n - i_2 = 2m_2^\ell, \;\; n - i_3 = 4m_3^\ell.$$

We claim that $(n - i_2)^2 \neq (n - i_1)(n - i_3)$. If not, put $b = n - i_2$ and $n - i_1 = b - x$, $n - i_3 = b + y$, where $0 < |x|, |y| < k$. Hence

$$b^2 = (b - x)(b + y) \quad \text{ or } \quad (y - x)b = xy,$$

where $x = y$ is plainly impossible. Now we have by part **(1)**

$$|xy| \;=\; b|y - x| \;\geq\; b \;>\; n - k \;>\; (k - 1)^2 \;\geq\; |xy|,$$

which is absurd.

So we have $m_2^2 \neq m_1 m_3$, where we assume $m_2^2 > m_1 m_3$ (the other case being analogous), and proceed to our last chains of inequalities. We obtain

$$\begin{aligned} 2(k-1)n \;&>\; n^2 - (n-k+1)^2 \;>\; (n-i_2)^2 - (n-i_1)(n-i_3) \\ &=\; 4[m_2^{2\ell} - (m_1 m_3)^\ell] \;\geq\; 4[(m_1 m_3 + 1)^\ell - (m_1 m_3)^\ell] \\ &\geq\; 4\ell m_1^{\ell-1} m_3^{\ell-1}. \end{aligned}$$

Since $\ell \geq 3$ and $n > k^\ell \geq k^3 > 6k$, this yields

$$\begin{aligned} 2(k-1)nm_1m_3 \;&>\; 4\ell m_1^\ell m_3^\ell \;=\; \ell(n-i_1)(n-i_3) \\ &>\; \ell(n-k+1)^2 \;>\; 3(n - \frac{n}{6})^2 \;>\; 2n^2. \end{aligned}$$

Now since $m_i \leq n^{1/\ell} \leq n^{1/3}$ we finally obtain

$$kn^{2/3} \;\geq\; km_1m_3 \;>\; (k-1)m_1m_3 \;>\; n,$$

or $k^3 > n$. With this contradiction, the proof is complete. $\square$

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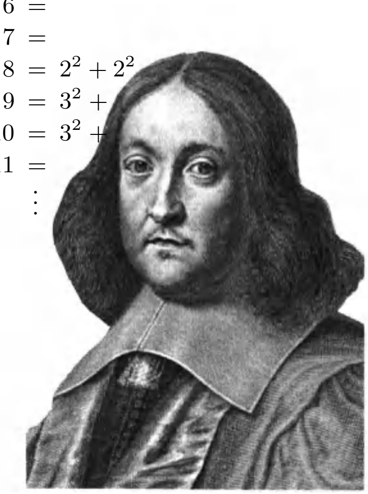
Representing numbers as sums of two squares

Chapter 4

$$\begin{array}{rcl} 1 &=& 1^2+0^2 \\ 2 &=& 1^2+1^2 \\ 3 &=& \\ 4 &=& 2^2+0^2 \\ 5 &=& 2^2+1^2 \\ 6 &=& \\ 7 &=& \\ 8 &=& 2^2+2^2 \\ 9 &=& 3^2+ \\ 10 &=& 3^2+ \\ 11 &=& \\ &\vdots& \end{array}$$



Pierre de Fermat

Which numbers can be written as sums of two squares?

This question is as old as number theory, and its solution is a classic in the field. The "hard" part of the solution is to see that every prime number of the form $4m+1$ is a sum of two squares. G. H. Hardy writes that this *two square theorem* of Fermat "is ranked, very justly, as one of the finest in arithmetic." Nevertheless, one of our Book Proofs below is quite recent.

Let's start with some "warm-ups." First, we need to distinguish between the prime $p=2$, the primes of the form $p=4m+1$, and the primes of the form $p=4m+3$. Every prime number belongs to exactly one of these three classes. At this point we may note (using a method "à la Euclid") that there are infinitely many primes of the form $4m+3$. In fact, if there were only finitely many, then we could take p_k to be the largest prime of this form. Setting

$$N_k \;:=\; 2^2\cdot 3\cdot 5\cdots p_k - 1$$

(where $p_1=2, p_2=3, p_3=5,\dots$ denotes the sequence of all primes), we find that N_k is congruent to $3 \pmod 4$, so it must have a prime factor of the form $4m+3$, and this prime factor is larger than p_k — contradiction.

Our first lemma characterizes the primes for which -1 is a square in the field $\mathbb{Z}_p$ (which is reviewed in the box on the next page). It will also give us a quick way to derive that there are infinitely many primes of the form $4m+1$.

Lemma 1. *For primes $p=4m+1$ the equation $s^2\equiv -1 \pmod p$ has two solutions $s\in\{1,2,\dots,p{-}1\}$, for $p=2$ there is one such solution, while for primes of the form $p=4m+3$ there is no solution.*

■ **Proof.** For $p=2$ take $s=1$. For odd p, we construct the equivalence relation on $\{1,2,\dots,p-1\}$ that is generated by identifying every element with its additive inverse and with its multiplicative inverse in $\mathbb{Z}_p$. Thus the "general" equivalence classes will contain four elements

$$\{x,-x,\overline{x},-\overline{x}\}$$

since such a 4-element set contains both inverses for all its elements. However, there are smaller equivalence classes if some of the four numbers are not distinct:

- $x \equiv -x$ is impossible for odd p.
- $x \equiv \overline{x}$ is equivalent to $x^2 \equiv 1$. This has two solutions, namely $x = 1$ and $x = p - 1$, leading to the equivalence class $\{1, p-1\}$ of size 2.
- $x \equiv -\overline{x}$ is equivalent to $x^2 \equiv -1$. This equation may have no solution or two distinct solutions $x_0, p - x_0$: in this case the equivalence class is $\{x_0, p - x_0\}$.

For $p = 11$ the partition is $\{1, 10\}, \{2, 9, 6, 5\}, \{3, 8, 4, 7\}$; for $p = 13$ it is $\{1, 12\}, \{2, 11, 7, 6\}, \{3, 10, 9, 4\}, \{5, 8\}$: the pair $\{5, 8\}$ yields the two solutions of $s^2 \equiv -1 \bmod 13$.

The set $\{1, 2, \ldots, p-1\}$ has $p-1$ elements, and we have partitioned it into quadruples (equivalence classes of size 4), plus one or two pairs (equivalence classes of size 2). For $p - 1 = 4m + 2$ we find that there is only the one pair $\{1, p-1\}$, the rest is quadruples, and thus $s^2 \equiv -1 \pmod{p}$ has no solution. For $p - 1 = 4m$ there has to be the second pair, and this contains the two solutions of $s^2 \equiv -1$ that we were looking for. □

Lemma 1 says that every odd prime dividing a number $M^2 + 1$ must be of the form $4m + 1$. This implies that there are infinitely many primes of this form: Otherwise, look at $(2 \cdot 3 \cdot 5 \cdots q_k)^2 + 1$, where q_k is the largest such prime. The same reasoning as above yields a contradiction.

Prime fields

If p is a prime, then the set $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ with addition and multiplication defined "modulo p" forms a finite field. We will need the following simple properties:

- For $x \in \mathbb{Z}_p$, $x \neq 0$, the additive inverse (for which we usually write $-x$) is given by $p - x \in \{1, 2, \ldots, p-1\}$. If $p > 2$, then x and $-x$ are different elements of $\mathbb{Z}_p$.
- Each $x \in \mathbb{Z}_p \backslash \{0\}$ has a unique multiplicative inverse $\overline{x} \in \mathbb{Z}_p \backslash \{0\}$, with $x\overline{x} \equiv 1 \pmod{p}$.
  The definition of primes implies that the map $\mathbb{Z}_p \to \mathbb{Z}_p$, $z \mapsto xz$ is injective for $x \neq 0$. Thus on the finite set $\mathbb{Z}_p \backslash \{0\}$ it must be surjective as well, and hence for each x there is a unique $\overline{x} \neq 0$ with $x\overline{x} \equiv 1 \pmod{p}$.
- The squares $0^2, 1^2, 2^2, \ldots, h^2$ define different elements of $\mathbb{Z}_p$, for $h = \lfloor \frac{p}{2} \rfloor$.
  This is since $x^2 \equiv y^2$, or $(x+y)(x-y) \equiv 0$, implies that $x \equiv y$ or that $x \equiv -y$. The $1 + \lfloor \frac{p}{2} \rfloor$ elements $0^2, 1^2, \ldots, h^2$ are called the *squares* in $\mathbb{Z}_p$.

$+$	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

$\cdot$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Addition and multiplication in $\mathbb{Z}_5$

At this point, let us note "on the fly" that for *all* primes there are solutions for $x^2 + y^2 \equiv -1 \pmod{p}$. In fact, there are $\lfloor \frac{p}{2} \rfloor + 1$ distinct squares x^2 in $\mathbb{Z}_p$, and there are $\lfloor \frac{p}{2} \rfloor + 1$ distinct numbers of the form $-(1 + y^2)$. These two sets of numbers are too large to be disjoint, since $\mathbb{Z}_p$ has only p elements, and thus there must exist x and y with $x^2 \equiv -(1 + y^2) \pmod{p}$.

Lemma 2. *No number $n = 4m + 3$ is a sum of two squares.*

■ **Proof.** The square of any even number is $(2k)^2 = 4k^2 \equiv 0 \pmod 4$, while squares of odd numbers yield $(2k+1)^2 = 4(k^2+k)+1 \equiv 1 \pmod 4$. Thus any sum of two squares is congruent to $0, 1$ or $2 \pmod 4$. □

This is enough evidence for us that the primes $p = 4m+3$ are "bad." Thus, we proceed with "good" properties for primes of the form $p = 4m+1$. On the way to the main theorem, the following is the key step.

Proposition. *Every prime of the form $p = 4m+1$ is a sum of two squares, that is, it can be written as $p = x^2+y^2$ for some natural numbers $x, y \in \mathbb{N}$.*

We shall present here two proofs of this result — both of them elegant and surprising. The first proof features a striking application of the "pigeon-hole principle" (which we have already used "on the fly" before Lemma 2; see Chapter 25 for more), as well as a clever move to arguments "modulo p" and back. The idea is due to the Norwegian number theorist Axel Thue.

■ **Proof.** Consider the pairs (x', y') of integers with $0 \le x', y' \le \sqrt{p}$, that is, $x', y' \in \{0, 1, \ldots, \lfloor\sqrt{p}\rfloor\}$. There are $(\lfloor\sqrt{p}\rfloor + 1)^2$ such pairs. Using the estimate $\lfloor x\rfloor + 1 > x$ for $x = \sqrt{p}$, we see that we have more than p such pairs of integers. Thus for any $s \in \mathbb{Z}$, it is impossible that all the values $x' - sy'$ produced by the pairs (x', y') are distinct modulo p. That is, for every s there are two distinct pairs

$$(x', y'),\ (x'', y'') \ \in \ \{0, 1, \ldots, \lfloor\sqrt{p}\rfloor\}^2$$

with $x' - sy' \equiv x'' - sy'' \pmod p$. Now we take differences: We have $x' - x'' \equiv s(y' - y'') \pmod p$. Thus if we define $x := |x' - x''|$, $y := |y' - y''|$, then we get

$$(x, y) \in \{0, 1, \ldots, \lfloor\sqrt{p}\rfloor\}^2 \quad \text{with} \quad x \equiv \pm sy \pmod p.$$

Also we know that not both x and y can be zero, because the pairs (x', y') and (x'', y'') are distinct.

Now let s be a solution of $s^2 \equiv -1 \pmod p$, which exists by Lemma 1. Then $x^2 \equiv s^2y^2 \equiv -y^2 \pmod p$, and so we have produced

$$(x, y) \in \mathbb{Z}^2 \quad \text{with} \quad 0 < x^2 + y^2 < 2p \quad \text{and} \quad x^2 + y^2 \equiv 0 \pmod p.$$

But p is the only number between 0 and $2p$ that is divisible by p. Thus $x^2 + y^2 = p$: done! □

For $p = 13$, $\lfloor\sqrt{p}\rfloor = 3$ we consider $x', y' \in \{0, 1, 2, 3\}$. For $s = 5$, the sum $x' - sy' \pmod{13}$ assumes the following values:

$x' \backslash y'$	0	1	2	3
0	0	8	3	11
1	1	9	4	12
2	2	10	5	0
3	3	11	6	1

Our second proof for the proposition — also clearly a Book Proof — was discovered by Roger Heath-Brown in 1971 and appeared in 1984. (A condensed "one-sentence version" was given by Don Zagier.) It is so elementary that we don't even need to use Lemma 1.

Heath-Brown's argument features three linear involutions: a quite obvious one, a hidden one, and a trivial one that gives "the final blow." The second, unexpected, involution corresponds to some hidden structure on the set of integral solutions of the equation $4xy + z^2 = p$.

■ **Proof.** We study the set

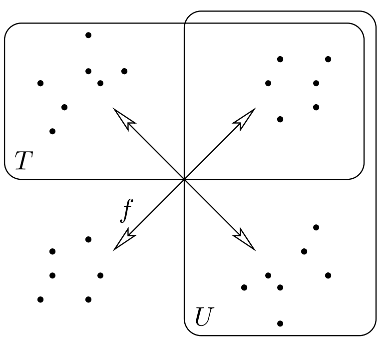
$$S \quad := \quad \{(x, y, z) \in \mathbb{Z}^3 : 4xy + z^2 = p, \quad x > 0, \quad y > 0\}.$$

This set is finite. Indeed, $x \geq 1$ and $y \geq 1$ implies $y \leq \frac{p}{4}$ and $x \leq \frac{p}{4}$. So there are only finitely many possible values for x and y, and given x and y, there are at most two values for z.

1. The first linear involution is given by

$$f : S \longrightarrow S, \quad (x, y, z) \longmapsto (y, x, -z),$$

that is, "interchange x and y, and negate z." This clearly maps S to itself, and it is an *involution*: Applied twice, it yields the identity. Also, f has no fixed points, since $z = 0$ would imply $p = 4xy$, which is impossible. Furthermore, f maps the solutions in



$$T \quad := \quad \{(x, y, z) \in S : z > 0\}$$

to the solutions in $S \backslash T$, which satisfy $z < 0$. Also, f reverses the signs of $x - y$ and of z, so it maps the solutions in

$$U \quad := \quad \{(x, y, z) \in S : (x - y) + z > 0\}$$

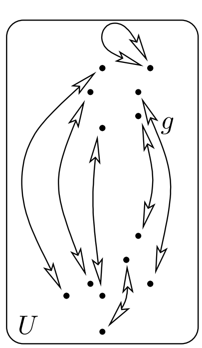
to the solutions in $S \backslash U$. For this we have to see that there is no solution with $(x-y)+z = 0$, but there is none since this would give $p = 4xy+z^2 = 4xy + (x - y)^2 = (x + y)^2$.

What do we get from the study of f? The main observation is that since f maps the sets T and U to their complements, it also interchanges the elements in $T \backslash U$ with these in $U \backslash T$. That is, there is the same number of solutions in U that are not in T as there are solutions in T that are not in U — *so T and U have the same cardinality.*

2. The second involution that we study is an involution on the set U:

$$g : U \longrightarrow U, \quad (x, y, z) \longmapsto (x - y + z, y, 2y - z).$$

First we check that indeed this is a well-defined map: If $(x, y, z) \in U$, then $x - y + z > 0, y > 0$ and $4(x - y + z)y + (2y - z)^2 = 4xy + z^2$, so $g(x, y, z) \in S$. By $(x - y + z) - y + (2y - z) = x > 0$ we find that indeed $g(x, y, z) \in U$.



Also g is an involution: $g(x, y, z) = (x - y + z, y, 2y - z)$ is mapped by g to $((x - y + z) - y + (2y - z), y, 2y - (2y - z)) = (x, y, z)$.

And finally: g has exactly one fixed point:

$$(x, y, z) \quad = \quad g(x, y, z) \quad = \quad (x - y + z, y, 2y - z)$$

holds exactly if $y = z$: But then $p = 4xy + y^2 = (4x + y)y$, which holds only for $y = 1 = z$, and $x = \frac{p-1}{4}$.

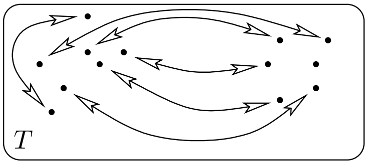
But if g is an involution on U that has exactly one fixed point, then *the cardinality of U is odd.*

3. The third, trivial, involution that we study is the involution on T that interchanges x and y:

$$h : T \longrightarrow T, \quad (x, y, z) \longmapsto (y, x, z).$$

This map is clearly well-defined, and an involution. We combine now our knowledge derived from the other two involutions: The cardinality of T is equal to the cardinality of U, which is odd. But if h is an involution on a finite set of odd cardinality, then it *has a fixed point*: There is a point $(x, y, z) \in T$ with $x = y$, that is, a solution of

$$p = 4x^2 + z^2 = (2x)^2 + z^2. \qquad \square$$



On a finite set of odd cardinality, every involution has at least one fixed point.

Note that this proof yields more — the number of representations of p in the form $p = x^2 + (2y)^2$ is *odd* for all primes of the form $p = 4m+1$. (The representation is actually unique, see [3].) Also note that both proofs are not effective: Try to find x and y for a ten digit prime! Efficient ways to find such representations as sums of two squares are discussed in [1] and [7].

The following theorem completely answers the question which started this chapter.

Theorem. *A natural number n can be represented as a sum of two squares if and only if every prime factor of the form $p = 4m + 3$ appears with an even exponent in the prime decomposition of n.*

■ **Proof.** Call a number n *representable* if it is a sum of two squares, that is, if $n = x^2 + y^2$ for some $x, y \in \mathbb{N}_0$. The theorem is a consequence of the following five facts.

(1) $1 = 1^2 + 0^2$ and $2 = 1^2 + 1^2$ are representable. Every prime of the form $p = 4m + 1$ is representable.

(2) The product of any two representable numbers $n_1 = x_1^2 + y_1^2$ and $n_2 = x_2^2 + y_2^2$ is representable: $n_1 n_2 = (x_1 x_2 + y_1 y_2)^2 + (x_1 y_2 - x_2 y_1)^2$.

(3) If n is representable, $n = x^2 + y^2$, then also nz^2 is representable, by $nz^2 = (xz)^2 + (yz)^2$.

Facts (1), (2) and (3) together yield the "if" part of the theorem.

(4) If $p = 4m + 3$ is a prime that divides a representable number $n = x^2 + y^2$, then p divides both x and y, and thus p^2 divides n. In fact, if we had $x \not\equiv 0 \pmod{p}$, then we could find $\overline{x}$ such that $x\overline{x} \equiv 1 \pmod{p}$, multiply the equation $x^2 + y^2 \equiv 0$ by $\overline{x}^2$, and thus obtain $1 + y^2\overline{x}^2 = 1 + (\overline{x}y)^2 \equiv 0 \pmod{p}$, which is impossible for $p = 4m + 3$ by Lemma 1.

(5) If n is representable, and $p = 4m + 3$ divides n, then p^2 divides n, and n/p^2 is representable. This follows from (4), and completes the proof. $\square$

Two remarks close our discussion:

- If a and b are two natural numbers that are relatively prime, then there are infinitely many primes of the form $am+b$ $(m \in \mathbb{N})$ — this is a famous (and difficult) theorem of Dirichlet. More precisely, one can show that the number of primes $p \leq x$ of the form $p = am + b$ is described very accurately for large x by the function $\frac{1}{\varphi(a)} \frac{x}{\log x}$, where $\varphi(a)$ denotes the number of b with $1 \leq b < a$ that are relatively prime to a. (This is a substantial refinement of the prime number theorem, which we had discussed on page 10.)
- This means that the primes for fixed a and varying b appear essentially at the same rate. Nevertheless, for example for $a = 4$ one can observe a rather subtle, but still noticeable and persistent tendency towards "more" primes of the form $4m+3$: If you look for a large random x, then chances are that there are more primes $p \leq x$ of the form $p = 4m + 3$ than of the form $p = 4m + 1$. This effect is known as "Chebyshev's bias"; see Riesel [4] and Rubinstein and Sarnak [5].

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The law of quadratic reciprocity

Chapter 5

Carl Friedrich Gauss

Which famous mathematical theorem has been proved most often? Pythagoras would certainly be a good candidate or the fundamental theorem of algebra, but the champion is without doubt the law of quadratic reciprocity in number theory. In an admirable monograph Franz Lemmermeyer lists as of the year 2000 no fewer than 196 proofs. Many of them are of course only slight variations of others, but the array of different ideas is still impressive, as is the list of contributors. Carl Friedrich Gauss gave the first complete proof in 1801 and followed up with seven more. A little later Ferdinand Gotthold Eisenstein added five more — and the ongoing list of provers reads like a Who is Who of mathematics.

With so many proofs present the question which of them belongs in the Book can have no easy answer. Is it the shortest, the most unexpected, or should one look for the proof that had the greatest potential for generalizations to other and deeper reciprocity laws? We have chosen two proofs (based on Gauss' third and sixth proofs), of which the first may be the simplest and most pleasing, while the other is the starting point for fundamental results in more general structures.

As in the previous chapter we work "modulo p", where p is an odd prime; $\mathbb{Z}_p$ is the field of residues upon division by p, and we usually (but not always) take these residues as $0, 1, \ldots, p-1$. Consider some $a \not\equiv 0 \pmod{p}$, that is, $p \nmid a$. We call a a *quadratic residue* modulo p if $a \equiv b^2 \pmod{p}$ for some b, and a *quadratic nonresidue* otherwise. The quadratic residues are therefore $1^2, 2^2, \ldots, (\frac{p-1}{2})^2$, and so there are $\frac{p-1}{2}$ quadratic residues and $\frac{p-1}{2}$ quadratic nonresidues. Indeed, if $i^2 \equiv j^2 \pmod{p}$ with $1 \le i, j \le \frac{p-1}{2}$, then $p \mid i^2 - j^2 = (i-j)(i+j)$. As $2 \le i + j \le p-1$ we have $p \mid i - j$, that is, $i \equiv j \pmod{p}$.

For $p = 13$, the quadratic residues are $1^2 \equiv 1$, $2^2 \equiv 4$, $3^2 \equiv 9$, $4^2 \equiv 3$, $5^2 \equiv 12$, and $6^2 \equiv 10$; the nonresidues are $2, 5, 6, 7, 8, 11$.

At this point it is convenient to introduce the so-called *Legendre symbol*. Let $a \not\equiv 0 \pmod{p}$, then

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } a \text{ is a quadratic residue,} \\ -1 & \text{if } a \text{ is a quadratic nonresidue.} \end{cases}$$

The story begins with Fermat's "little theorem": For $a \not\equiv 0 \pmod{p}$,

$$a^{p-1} \equiv 1 \pmod{p}. \tag{1}$$

In fact, since $\mathbb{Z}_p^* = \mathbb{Z}_p \setminus \{0\}$ is a group with multiplication, the set $\{1a, 2a, 3a, \ldots, (p-1)a\}$ runs again through all nonzero residues,

$$(1a)(2a)\cdots((p-1)a) \equiv 1 \cdot 2 \cdots (p-1) \pmod{p},$$

and hence by dividing by $(p-1)!$, we get $a^{p-1} \equiv 1 \pmod{p}$.

In other words, the *polynomial* $x^{p-1} - 1 \in \mathbb{Z}_p[x]$ has as roots all nonzero residues. Next we note that

$$x^{p-1} - 1 = (x^{\frac{p-1}{2}} - 1)(x^{\frac{p-1}{2}} + 1).$$

Suppose $a \equiv b^2 \pmod p$ is a quadratic residue. Then by Fermat's little theorem $a^{\frac{p-1}{2}} \equiv b^{p-1} \equiv 1 \pmod p$. Hence the quadratic residues are precisely the roots of the first factor $x^{\frac{p-1}{2}} - 1$, and the $\frac{p-1}{2}$ nonresidues must therefore be the roots of the second factor $x^{\frac{p-1}{2}} + 1$. Comparing this to the definition of the Legendre symbol, we obtain the following important tool.

For example, for $p = 17$ and $a = 3$ we have $3^8 = (3^4)^2 = 81^2 \equiv (-4)^2 \equiv -1 \pmod{17}$, while for $a = 2$ we get $2^8 = (2^4)^2 \equiv (-1)^2 \equiv 1 \pmod{17}$. Hence 2 is a quadratic residue, while 3 is a nonresidue.

Euler's criterion. *For $a \not\equiv 0 \pmod p$,*

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod p.$$

This gives us at once the important *product rule*

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right), \tag{2}$$

since this obviously holds for the right-hand side of Euler's criterion. The product rule is extremely helpful when one tries to compute Legendre symbols: Since any integer is a product of ± 1 and primes we only have to compute $(\frac{-1}{p})$, $(\frac{2}{p})$, and $(\frac{q}{p})$ for odd primes q.

By Euler's criterion $(\frac{-1}{p}) = 1$ if $p \equiv 1 \pmod 4$, and $(\frac{-1}{p}) = -1$ if $p \equiv 3 \pmod 4$, something we have already seen in the previous chapter. The case $(\frac{2}{p})$ will follow from the Lemma of Gauss below: $(\frac{2}{p}) = 1$ if $p \equiv \pm 1 \pmod 8$, while $(\frac{2}{p}) = -1$ if $p \equiv \pm 3 \pmod 8$.

Gauss did lots of calculations with quadratic residues and, in particular, studied whether there might be a relation between q being a quadratic residue modulo p and p being a quadratic residue modulo q, when p and q are odd primes. After a lot of experimentation he conjectured and then proved the following theorem.

Law of quadratic reciprocity. *Let p and q be different odd primes. Then*

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

If $p \equiv 1 \pmod 4$ or $q \equiv 1 \pmod 4$, then $\frac{p-1}{2}$ (resp. $\frac{q-1}{2}$) is even, and therefore $(-1)^{\frac{p-1}{2}\frac{q-1}{2}} = 1$; thus $(\frac{q}{p}) = (\frac{p}{q})$. When $p \equiv q \equiv 3 \pmod 4$, we have $(\frac{p}{q}) = -(\frac{q}{p})$. Thus for odd primes we get $(\frac{p}{q}) = (\frac{q}{p})$ unless *both* p and q are congruent to $3 \pmod 4$.

Example: $(\frac{3}{17}) = (\frac{17}{3}) = (\frac{2}{3}) = -1$, so 3 is a nonresidue mod 17.

First proof. The key to our first proof (which is Gauss' third) is a counting formula that soon came to be called the *Lemma of Gauss*:

> **Lemma of Gauss.** *Suppose $a \not\equiv 0 \pmod p$. Take the numbers $1a, 2a, \dots, \frac{p-1}{2}a$ and reduce them modulo p to the residue system smallest in absolute value, $ia \equiv r_i \pmod p$ with $-\frac{p-1}{2} \le r_i \le \frac{p-1}{2}$ for all i. Then*
>
> $$\left(\frac{a}{p}\right) = (-1)^s, \quad \textit{where } s = \#\{i : r_i < 0\}.$$

■ **Proof.** Suppose $u_1, \dots, u_s$ are the residues smaller than 0, and that $v_1, \dots, v_{\frac{p-1}{2}-s}$ are those greater than 0. Then the numbers $-u_1, \dots, -u_s$ are between 1 and $\frac{p-1}{2}$, and are all different from the v_js (see the margin); hence $\{-u_1, \dots, -u_s, v_1, \dots, v_{\frac{p-1}{2}-s}\} = \{1, 2, \dots, \frac{p-1}{2}\}$. Therefore

If $-u_i = v_j$, then $u_i + v_j \equiv 0 \pmod p$. Now $u_i \equiv ka$, $v_j \equiv \ell a \pmod p$ implies $p \mid (k+\ell)a$. As p and a are relatively prime, p must divide $k + \ell$ which is impossible, since $k + \ell \le p - 1$.

$$\prod_i (-u_i) \prod_j v_j = \tfrac{p-1}{2}!,$$

which implies

$$(-1)^s \prod_i u_i \prod_j v_j \;\equiv\; \tfrac{p-1}{2}! \pmod p.$$

Now remember how we obtained the numbers u_i and v_j; they are the residues of $1a, \cdots, \frac{p-1}{2}a$. Hence

$$\tfrac{p-1}{2}! \;\equiv\; (-1)^s \prod_i u_i \prod_j v_j \;\equiv\; (-1)^s\, \tfrac{p-1}{2}!\, a^{\frac{p-1}{2}} \pmod p.$$

Cancelling $\frac{p-1}{2}!$ together with Euler's criterion gives

$$\left(\frac{a}{p}\right) \;\equiv\; a^{\frac{p-1}{2}} \;\equiv\; (-1)^s \pmod p,$$

and therefore $\left(\frac{a}{p}\right) = (-1)^s$, since p is odd. □

With this we can easily compute $\left(\frac{2}{p}\right)$: Since $1 \cdot 2, 2 \cdot 2, \dots, \frac{p-1}{2} \cdot 2$ are all between 1 and $p-1$, we have

$$s \;=\; \#\{i : \tfrac{p-1}{2} < 2i \le p-1\} \;=\; \tfrac{p-1}{2} - \#\{i : 2i \le \tfrac{p-1}{2}\} \;=\; \lceil \tfrac{p-1}{4} \rceil.$$

Check that s is even precisely for $p = 8k \pm 1$.

The Lemma of Gauss is the basis for many of the published proofs of the quadratic reciprocity law. The most elegant may be the one suggested by Ferdinand Gotthold Eisenstein, who had learned number theory from Gauss' famous *Disquisitiones Arithmeticae* and made important contributions to "higher reciprocity theorems" before his premature death at age 29. His proof is just counting lattice points!

Let p and q be odd primes, and consider $(\frac{q}{p})$. Suppose iq is a multiple of q that reduces to a negative residue $r_i < 0$ in the Lemma of Gauss. This means that there is a unique integer j such that $-\frac{p}{2} < iq - jp < 0$. Note that $0 < j < \frac{q}{2}$ since $0 < i < \frac{p}{2}$. In other words, $(\frac{q}{p}) = (-1)^s$, where s is the number of lattice points (x, y), that is, pairs of integers x, y satisfying

$$0 < py - qx < \frac{p}{2},\ 0 < x < \frac{p}{2},\ 0 < y < \frac{q}{2}. \tag{3}$$

Similarly, $(\frac{p}{q}) = (-1)^t$ where t is the number of lattice points (x, y) with

$$0 < qx - py < \frac{q}{2},\ 0 < x < \frac{p}{2},\ 0 < y < \frac{q}{2}. \tag{4}$$

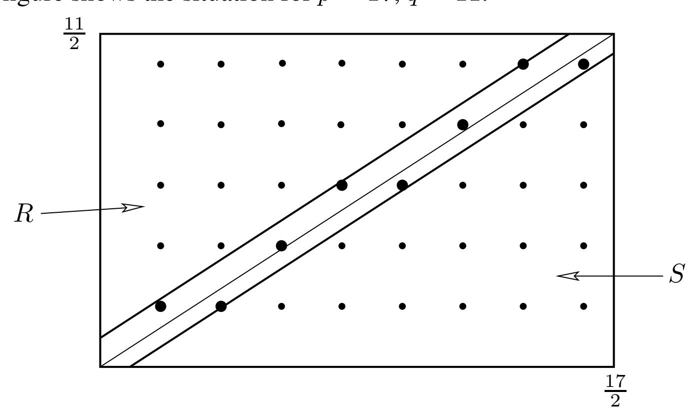
Now look at the rectangle with side lengths $\frac{p}{2}, \frac{q}{2}$, and draw the two lines parallel to the diagonal $py = qx$, $y = \frac{q}{p}x + \frac{1}{2}$ or $py - qx = \frac{p}{2}$, respectively, $y = \frac{q}{p}(x - \frac{1}{2})$ or $qx - py = \frac{q}{2}$.
The figure shows the situation for $p = 17$, $q = 11$.

$p = 17 \quad q = 11$

$s = 5 \quad t = 3$

$\left(\frac{q}{p}\right) = (-1)^5 = -1$

$\left(\frac{p}{q}\right) = (-1)^3 = -1$



The proof is now quickly completed by the following three observations:

1. There are no lattice points on the diagonal and the two parallels. This is so because $py = qx$ would imply $p \mid x$, which cannot be. For the parallels observe that $py - qx$ is an integer while $\frac{p}{2}$ or $\frac{q}{2}$ are not.

2. The lattice points observing (3) are precisely the points in the upper strip $0 < py - qx < \frac{p}{2}$, and those of (4) the points in the lower strip $0 < qx - py < \frac{q}{2}$. Hence the number of lattice points in the two strips is $s + t$.

3. The outer regions $R : py - qx > \frac{p}{2}$ and $S : qx - py > \frac{q}{2}$ contain the *same* number of points. To see this consider the map $\varphi : R \to S$ which maps (x, y) to $(\frac{p+1}{2} - x, \frac{q+1}{2} - y)$ and check that φ is an involution.

Since the total number of lattice points in the rectangle is $\frac{p-1}{2} \cdot \frac{q-1}{2}$, we infer that $s + t$ and $\frac{p-1}{2} \cdot \frac{q-1}{2}$ have the same parity, and so

$$\Big(\frac{q}{p}\Big)\Big(\frac{p}{q}\Big) \;=\; (-1)^{s+t} = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}. \qquad \square$$

Second proof. Our second choice does not use Gauss' lemma, instead it employs so-called "Gauss sums" in finite fields. Gauss invented them in his study of the equation $x^p - 1 = 0$ and the arithmetical properties of the field $\mathbb{Q}(\zeta)$ (called cyclotomic field), where ζ is a p-th root of unity. They have been the starting point for the search for higher reciprocity laws in general number fields.

Let us first collect a few facts about finite fields.

A. Let p and q be different odd primes, and consider the finite field F with q^{p-1} elements. Its prime field is $\mathbb{Z}_q$, whence $qa = 0$ for any $a \in F$. This implies that $(a+b)^q = a^q + b^q$, since any binomial coefficient $\binom{q}{i}$ is a multiple of q for $0 < i < q$, and thus 0 in F. Note that Euler's criterion is an *equation* $(\frac{p}{q}) = p^{\frac{q-1}{2}}$ in the prime field $\mathbb{Z}_q$.

B. The multiplicative group $F^* = F \setminus \{0\}$ is cyclic of size $q^{p-1} - 1$ (see the box on the next page). Since by Fermat's little theorem p is a divisor of $q^{p-1} - 1$, there exists an element $\zeta \in F$ of order p, that is, $\zeta^p = 1$, and ζ generates the subgroup $\{\zeta, \zeta^2, \ldots, \zeta^p = 1\}$ of F^*. Note that any ζ^i $(i \neq p)$ is again a generator. Hence we obtain the polynomial decomposition $x^p - 1 = (x-\zeta)(x-\zeta^2)\cdots(x-\zeta^p)$.

Now we can go to work. Consider the *Gauss sum*

$$G := \sum_{i=1}^{p-1} \left(\frac{i}{p}\right) \zeta^i \in F,$$

where $(\frac{i}{p})$ is the Legendre symbol. For the proof we derive two different expressions for G^q and then set them equal.

First expression. We have

$$G^q = \sum_{i=1}^{p-1} \left(\frac{i}{p}\right)^q \zeta^{iq} = \sum_{i=1}^{p-1} \left(\frac{i}{p}\right) \zeta^{iq} = \left(\frac{q}{p}\right) \sum_{i=1}^{p-1} \left(\frac{iq}{p}\right) \zeta^{iq} = \left(\frac{q}{p}\right) G, \quad (5)$$

where the first equality follows from $(a+b)^q = a^q + b^q$, the second uses that $(\frac{i}{p})^q = (\frac{i}{p})$ since q is odd, the third one is derived from (2), which yields $(\frac{i}{p}) = (\frac{q}{p})(\frac{iq}{p})$, and the last one holds since iq runs with i through all nonzero residues modulo p.

Example: Take $p = 3$, $q = 5$. Then $G = \zeta - \zeta^2$ and $G^5 = \zeta^5 - \zeta^{10} = \zeta^2 - \zeta = -(\zeta - \zeta^2) = -G$, corresponding to $(\frac{5}{3}) = (\frac{2}{3}) = -1$.

Second expression. Suppose we can prove

$$G^2 = (-1)^{\frac{p-1}{2}} p\,, \quad (6)$$

then we are quickly done. Indeed,

$$G^q = G(G^2)^{\frac{q-1}{2}} = G(-1)^{\frac{p-1}{2}\frac{q-1}{2}} p^{\frac{q-1}{2}} = G\left(\frac{p}{q}\right)(-1)^{\frac{p-1}{2}\frac{q-1}{2}}. \quad (7)$$

Equating the expressions in (5) and (7) and cancelling G, which is nonzero by (6), we find $(\frac{q}{p}) = (\frac{p}{q})(-1)^{\frac{p-1}{2}\frac{q-1}{2}}$, and thus

$$\left(\frac{q}{p}\right)\left(\frac{p}{q}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}.$$

The multiplicative group of a finite field is cyclic

Let F^* be the multiplicative group of the field F, with $|F^*| = n$. Writing ord(a) for the order of an element, that is, the smallest positive integer k such that $a^k = 1$, we want to find an element $a \in F^*$ with ord$(a) = n$. If ord$(b) = d$, then by Lagrange's theorem, d divides n (see the margin on page 4). Classifying the elements according to their order, we have

$$n = \sum_{d|n} \psi(d), \text{ where } \psi(d) = \#\{b \in F^* : \text{ord}(b) = d\}. \qquad (8)$$

If ord$(b) = d$, then every element b^i $(i = 1, \ldots, d)$ satisfies $(b^i)^d = 1$ and is therefore a root of the polynomial $x^d - 1$. But, since F is a field, $x^d - 1$ has at most d roots, and so the elements $b, b^2, \ldots, b^d = 1$ are precisely these roots. In particular, every element of order d is of the form b^i.

On the other hand, it is easily checked that ord$(b^i) = \frac{d}{(i,d)}$, where (i, d) denotes the greatest common divisor of i and d. Hence ord$(b^i) = d$ if and only if $(i, d) = 1$, that is, if i and d are relatively prime. Denoting *Euler's function* by $\varphi(d) = \#\{i : 1 \leq i \leq d, (i, d) = 1\}$, we thus have $\psi(d) = \varphi(d)$ whenever $\psi(d) > 0$. Looking at (8) we find

$$n = \sum_{d|n} \psi(d) \leq \sum_{d|n} \varphi(d)\,.$$

But, as we are going to show that

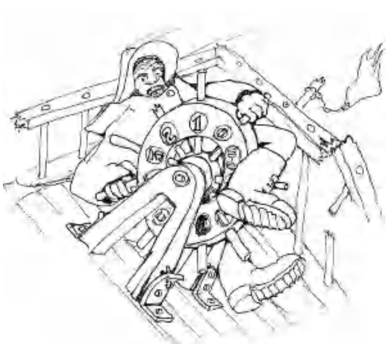
$$\sum_{d|n} \varphi(d) \;=\; n, \qquad (9)$$

we must have $\psi(d) = \varphi(d)$ for all d. In particular, $\psi(n) = \varphi(n) \geq 1$, and so there is an element of order n.

The following (folklore) proof of (9) belongs in the Book as well. Consider the n fractions

$$\tfrac{1}{n}, \tfrac{2}{n}, \ldots, \tfrac{k}{n}, \ldots, \tfrac{n}{n},$$

reduce them to lowest terms $\frac{k}{n} = \frac{i}{d}$ with $1 \leq i \leq d$, $(i, d) = 1$, $d \mid n$, and check that the denominator d appears precisely $\varphi(d)$ times.



*"Even in total chaos
we can hang on
to the cyclic group"*

It remains to verify (6), and for this we first make two simple observations:

- $\sum_{i=1}^{p} \zeta^i = 0$ and thus $\sum_{i=1}^{p-1} \zeta^i = -1$. Just note that $-\sum_{i=1}^{p} \zeta^i$ is the coefficient of x^{p-1} in $x^p - 1 = \prod_{i=1}^{p}(x - \zeta^i)$, and thus 0.
- $\sum_{k=1}^{p-1} (\frac{k}{p}) = 0$ and thus $\sum_{k=1}^{p-2} (\frac{k}{p}) = -(\frac{-1}{p})$, since there are equally many quadratic residues and nonresidues.

We have

$$G^2 = \Big(\sum_{i=1}^{p-1} \big(\frac{i}{p}\big)\,\zeta^i\Big)\Big(\sum_{j=1}^{p-1} \big(\frac{j}{p}\big)\,\zeta^j\Big) = \sum_{i,j} \big(\frac{ij}{p}\big)\,\zeta^{i+j}.$$

Setting $j \equiv ik \pmod p$ we find

$$G^2 = \sum_{i,k} \big(\frac{k}{p}\big)\,\zeta^{i(1+k)} = \sum_{k=1}^{p-1} \big(\frac{k}{p}\big) \sum_{i=1}^{p-1} \zeta^{(1+k)i}.$$

For $k = p-1 \equiv -1 \pmod p$ this gives $(\frac{-1}{p})(p-1)$, since $\zeta^{1+k} = 1$. Move $k = p-1$ in front and write

$$G^2 = \big(\frac{-1}{p}\big)(p-1) + \sum_{k=1}^{p-2} \big(\frac{k}{p}\big) \sum_{i=1}^{p-1} \zeta^{(1+k)i}.$$

Euler's criterion: $(\frac{-1}{p}) = (-1)^{\frac{p-1}{2}}$

Since ζ^{1+k} is a generator of the group for $k \neq p-1$, the inner sum equals $\sum_{i=1}^{p-1} \zeta^i = -1$ for all $k \neq p-1$ by our first observation. Hence the second summand is $-\sum_{k=1}^{p-2}(\frac{k}{p}) = (\frac{-1}{p})$ by our second observation. It follows that $G^2 = (\frac{-1}{p})p$ and thus with Euler's criterion $G^2 = (-1)^{\frac{p-1}{2}}p$, which completes the proof. $\square$

For $p = 3$, $q = 5$, $G^2 = (\zeta - \zeta^2)^2 = \zeta^2 - 2\zeta^3 + \zeta^4 = \zeta^2 - 2 + \zeta = -3 = (-1)^{\frac{3-1}{2}}3$, since $1 + \zeta + \zeta^2 = 0$.

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What's up?

I'm pushing 196 proofs
for quadratic reciprocity

Every finite division ring is a field

Chapter 6

Rings are important structures in modern algebra. If a ring R has a multiplicative unit element 1 and every nonzero element has a multiplicative inverse, then R is called a *division ring*. So, all that is missing in R from being a field is the commutativity of multiplication. The best-known example of a non-commutative division ring is the ring of quaternions discovered by Hamilton. But, as the chapter title says, every such division ring must of necessity be infinite. If R is finite, then the axioms force the multiplication to be commutative.

This result which is now a classic has caught the imagination of many mathematicians, because, as Herstein writes: "It is so unexpectedly interrelating two seemingly unrelated things, the number of elements in a certain algebraic system and the multiplication of that system."

Ernst Witt

Theorem. *Every finite division ring is commutative.*

This beautiful theorem which is usually attributed to MacLagan Wedderburn has been proved by many people using a variety of different ideas. Wedderburn himself gave three proofs in 1905, and another proof was given by Leonard E. Dickson in the same year. More proofs were later given by Emil Artin, Hans Zassenhaus, Nicolas Bourbaki, and many others. One proof stands out for its simplicity and elegance. It was found by Ernst Witt in 1931 and combines two elementary ideas towards a glorious finish.

■ **Proof.** Our first ingredient comes from a blend of linear algebra and basic group theory. For an arbitrary element $s \in R$, let C_s be the set $\{x \in R : xs = sx\}$ of elements which commute with s; C_s is called the *centralizer* of s. Clearly, C_s contains 0 and 1 and is a sub-division ring of R. The *center* Z is the set of elements which commute with all elements of R, thus $Z = \bigcap_{s \in R} C_s$. In particular, all elements of Z commute, 0 and 1 are in Z, and so Z is a *finite field*. Let us set $|Z| = q$.

We can regard R and C_s as vector spaces over the field Z and deduce that $|R| = q^n$, where n is the dimension of the vector space R over Z, and similarly $|C_s| = q^{n_s}$ for suitable integers $n_s \geq 1$.

Now let us assume that R is not a field. This means that for *some* $s \in R$ the centralizer C_s is not all of R, or, what is the same, $n_s < n$.

On the set $R^* := R \backslash \{0\}$ we consider the relation

$$r' \sim r \quad :\Longleftrightarrow \quad r' = x^{-1} r x \quad \text{for some } x \in R^*.$$

It is easy to check that $\sim$ is an equivalence relation. Let

$$A_s := \{x^{-1}sx : x \in R^*\}$$

be the equivalence class containing s. We note that $|A_s| = 1$ precisely when s is in the center Z. So by our assumption, there are classes A_s with $|A_s| \geq 2$. Consider now for $s \in R^*$ the map $f_s : x \longmapsto x^{-1}sx$ from R^* onto A_s. For $x, y \in R^*$ we find

$$\begin{aligned} x^{-1}sx = y^{-1}sy \quad &\Longleftrightarrow \quad (yx^{-1})s = s(yx^{-1}) \\ &\Longleftrightarrow \quad yx^{-1} \in C_s^* \quad \Longleftrightarrow \quad y \in C_s^* x, \end{aligned}$$

for $C_s^* := C_s \backslash \{0\}$, where $C_s^* x = \{zx : z \in C_s^*\}$ has size $|C_s^*|$. Hence any element $x^{-1}sx$ is the image of precisely $|C_s^*| = q^{n_s} - 1$ elements in R^* under the map f_s, and we deduce $|R^*| = |A_s|\,|C_s^*|$. In particular, we note that

$$\frac{|R^*|}{|C_s^*|} \;=\; \frac{q^n - 1}{q^{n_s} - 1} \;=\; |A_s| \quad \text{is an } \textit{integer} \text{ for all } s.$$

We know that the equivalence classes partition R^*. We now group the central elements Z^* together and denote by $A_1, \dots, A_t$ the equivalence classes containing more than one element. By our assumption we know $t \geq 1$. Since $|R^*| = |Z^*| + \sum_{k=1}^{t} |A_k|$, we have proved the so-called *class formula*

$$q^n - 1 \;=\; q - 1 + \sum_{k=1}^{t} \frac{q^n - 1}{q^{n_k} - 1}, \tag{1}$$

where we have $1 < \frac{q^n - 1}{q^{n_k} - 1} \in \mathbb{N}$ for all k.

With (1) we have left abstract algebra and are back to the natural numbers. Next we claim that $q^{n_k} - 1 \,|\, q^n - 1$ implies $n_k \,|\, n$. Indeed, write $n = an_k + r$ with $0 \leq r < n_k$, then $q^{n_k} - 1 \,|\, q^{an_k + r} - 1$ implies

$$q^{n_k} - 1 \,|\, (q^{an_k + r} - 1) - (q^{n_k} - 1) = q^{n_k}(q^{(a-1)n_k + r} - 1),$$

and thus $q^{n_k} - 1 \,|\, q^{(a-1)n_k + r} - 1$, since q^{n_k} and $q^{n_k} - 1$ are relatively prime. Continuing in this way we find $q^{n_k} - 1 \,|\, q^r - 1$ with $0 \leq r < n_k$, which is only possible for $r = 0$, that is, $n_k \,|\, n$. In summary, we note

$$n_k \,|\, n \quad \text{for all } k. \tag{2}$$

Now comes the second ingredient: the complex numbers $\mathbb{C}$. Consider the polynomial $x^n - 1$. Its roots in $\mathbb{C}$ are called the *n-th roots of unity*. Since $\lambda^n = 1$, all these roots λ have $|\lambda| = 1$ and lie therefore on the unit circle of the complex plane. In fact, they are precisely the numbers $\lambda_k = e^{\frac{2k\pi i}{n}} = \cos(2k\pi/n) + i\sin(2k\pi/n)$, $0 \leq k \leq n-1$ (see the box on the next page). Some of the roots λ satisfy $\lambda^d = 1$ for $d < n$; for example, the root $\lambda = -1$ satisfies $\lambda^2 = 1$. For a root λ, let d be the smallest positive exponent with $\lambda^d = 1$, that is, d is the order of λ in the group of the roots of unity. Then $d \,|\, n$, by Lagrange's theorem ("the order of every element of a group divides the order of the group" — see the box in Chapter 1). Note that there are roots of order n, such as $\lambda_1 = e^{\frac{2\pi i}{n}}$.

Roots of unity

Any complex number $z = x + iy$ may be written in the "polar" form

$$z \;=\; re^{i\varphi} \;=\; r(\cos\varphi + i\sin\varphi),$$

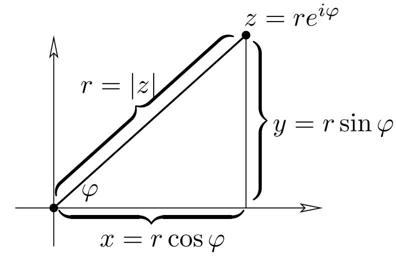
where $r = |z| = \sqrt{x^2+y^2}$ is the distance of z to the origin, and φ is the angle measured from the positive x-axis. The n-th roots of unity are therefore of the form

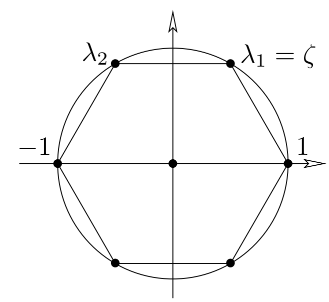
$$\lambda_k = e^{\frac{2k\pi i}{n}} = \cos(2k\pi/n) + i\sin(2k\pi/n), \qquad 0 \le k \le n-1,$$

since for all k

$$\lambda_k^n = e^{2k\pi i} = \cos(2k\pi) + i\sin(2k\pi) = 1.$$

We obtain these roots geometrically by inscribing a regular n-gon into the unit circle. Note that $\lambda_k = \zeta^k$ for all k, where $\zeta = e^{\frac{2\pi i}{n}}$. Thus the n-th roots of unity form a cyclic group $\{\zeta, \zeta^2, \ldots, \zeta^{n-1}, \zeta^n = 1\}$ of order n.





The roots of unity for $n = 6$

Now we group all roots of order d together and set

$$\phi_d(x) \;:=\; \prod_{\lambda \text{ of order } d} (x - \lambda).$$

Note that the definition of $\phi_d(x)$ is independent of n. Since every root has some order d, we conclude that

$$x^n - 1 \;=\; \prod_{d\,|\,n} \phi_d(x). \tag{3}$$

Here is the crucial observation: The *coefficients* of the polynomials $\phi_n(x)$ are *integers* (that is, $\phi_n(x) \in \mathbb{Z}[x]$ for all n), where in addition the constant coefficient is either 1 or -1.

Let us carefully verify this claim. For $n = 1$ we have 1 as the only root, and so $\phi_1(x) = x - 1$. Now we proceed by induction, where we assume $\phi_d(x) \in \mathbb{Z}[x]$ for all $d < n$, and that the constant coefficient of $\phi_d(x)$ is 1 or -1. By (3),

$$x^n - 1 \;=\; p(x)\,\phi_n(x) \tag{4}$$

where $p(x) = \sum\limits_{j=0}^{\ell} p_j x^j,\ \ \phi_n(x) = \sum\limits_{k=0}^{n-\ell} a_k x^k,$ with $p_0 = 1$ or $p_0 = -1$.

Since $-1 = p_0 a_0$, we see $a_0 \in \{1, -1\}$. Suppose we already know that $a_0, a_1, \ldots, a_{k-1} \in \mathbb{Z}$. Computing the coefficient of x^k on both sides of (4) we find

$$\sum_{j=0}^{k} p_j a_{k-j} \;=\; \sum_{j=1}^{k} p_j a_{k-j} + p_0 a_k \;\in\; \mathbb{Z}.$$

By assumption, all $a_0, \ldots, a_{k-1}$ (and all p_j) are in $\mathbb{Z}$. Thus $p_0 a_k$ and hence a_k must also be integers, since p_0 is 1 or -1.

We are ready for the *coup de grâce*. Let $n_k \,|\, n$ be one of the numbers appearing in (1). Then

$$x^n - 1 \;=\; \prod_{d\,|\,n} \phi_d(x) \;=\; (x^{n_k} - 1)\phi_n(x) \prod_{d\,|\,n,\, d\nmid n_k,\, d\neq n} \phi_d(x).$$

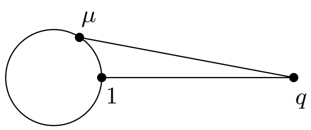
We conclude that in $\mathbb{Z}$ we have the divisibility relations

$$\phi_n(q) \mid q^n - 1 \qquad \text{and} \qquad \phi_n(q) \,\Big|\, \frac{q^n - 1}{q^{n_k} - 1}. \tag{5}$$

Since (5) holds for all k, we deduce from the class formula (1)

$$\phi_n(q) \,|\, q - 1,$$

but this cannot be. Why? We know $\phi_n(x) = \prod(x - \lambda)$ where λ runs through all roots of $x^n - 1$ of order n. Let $\widetilde{\lambda} = a + ib$ be one of those roots. By $n > 1$ (because of $R \neq Z$) we have $\widetilde{\lambda} \neq 1$, which implies that the real part a is smaller than 1. Now $|\widetilde{\lambda}|^2 = a^2 + b^2 = 1$, and hence



$|q - \mu| > |q - 1|$

$$\begin{aligned} |q - \widetilde{\lambda}|^2 &= |q - a - ib|^2 \;=\; (q-a)^2 + b^2 \\ &= q^2 - 2aq + a^2 + b^2 \;=\; q^2 - 2aq + 1 \\ &> q^2 - 2q + 1 \qquad \text{(because of } a < 1) \\ &= (q-1)^2, \end{aligned}$$

and so $|q - \widetilde{\lambda}| > q - 1$ holds for *all* roots of order n. This implies

$$|\phi_n(q)| \;=\; \prod_{\lambda} |q - \lambda| > q - 1,$$

which means that $\phi_n(q)$ cannot be a divisor of $q - 1$, contradiction and end of proof. $\square$

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Some irrational numbers — Chapter 7

> *“π is irrational”*

Charles Hermite

This was already conjectured by Aristotle, when he claimed that diameter and circumference of a circle are not commensurable. The first proof of this fundamental fact was given by Johann Heinrich Lambert in 1766. Our Book Proof is due to Ivan Niven, 1947: an extremely elegant one-page proof that needs only elementary calculus. Its idea is powerful, and quite a bit more can be derived from it, as was shown by Iwamoto and Koksma, respectively:

- π^2 is irrational and
- e^r is irrational for rational $r \neq 0$.

Niven’s method does, however, have its roots and predecessors: It can be traced back to the classical paper by Charles Hermite from 1873 which first established that e is transcendental, that is, that e is not a zero of a polynomial with rational coefficients.

Before we treat π we will look at e and its powers, and see that these are irrational. This is much easier, and we thus also follow the historical order in the development of the results.

$$\begin{aligned} e &:= 1+\tfrac{1}{1}+\tfrac{1}{2}+\tfrac{1}{6}+\tfrac{1}{24}+\dots \\ &= 2.718281828... \end{aligned}$$

$$\begin{aligned} e^x &:= 1+\tfrac{x}{1}+\tfrac{x^2}{2}+\tfrac{x^4}{6}+\tfrac{x^4}{24}+\dots \\ &= \sum_{k\geq 0}\tfrac{x^k}{k!} \end{aligned}$$

To start with, it is rather easy to see (as did Fourier in 1815) that $e = \sum_{k\geq 0}\frac{1}{k!}$ is irrational. Indeed, if we had $e = \frac{a}{b}$ for integers a and $b > 0$, then we would get

$$n!be \;=\; n!a$$

for *every* $n \geq 0$. But this cannot be true, because on the right-hand side we have an integer, while the left-hand side with

$$e \;=\; \Big(1+\frac{1}{1!}+\frac{1}{2!}+\ldots+\frac{1}{n!}\Big) + \Big(\frac{1}{(n+1)!}+\frac{1}{(n+2)!}+\frac{1}{(n+3)!}+\ldots\Big)$$

decomposes into an integral part

$$bn!\Big(1+\frac{1}{1!}+\frac{1}{2!}+\ldots+\frac{1}{n!}\Big)$$

and a second part

$$b\Big(\frac{1}{n+1}+\frac{1}{(n+1)(n+2)}+\frac{1}{(n+1)(n+2)(n+3)}+\;\ldots\Big)$$

Geometric series

For the infinite geometric series

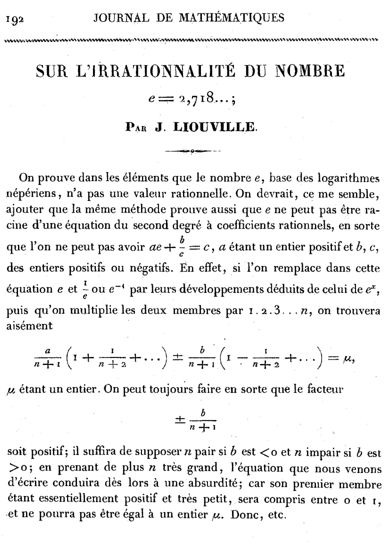
$Q = \frac{1}{q} + \frac{1}{q^2} + \frac{1}{q^3} + \ldots$

with $q > 1$ we clearly have

$qQ = 1 + \frac{1}{q} + \frac{1}{q^2} + \ldots = 1 + Q$

and thus

$Q = \dfrac{1}{q-1}.$



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SUR L'IRRATIONNALITÉ DU NOMBRE

$e = 2,718\ldots;$

PAR J. LIOUVILLE.

On prouve dans les éléments que le nombre e, base des logarithmes népériens, n'a pas une valeur rationnelle. On devrait, ce me semble, ajouter que la même méthode prouve aussi que e ne peut pas être racine d'une équation du second degré à coefficients rationnels, en sorte que l'on ne peut pas avoir $ae + \frac{b}{e} = c$, a étant un entier positif et b, c, des entiers positifs ou négatifs. En effet, si l'on remplace dans cette équation e et $\frac{1}{e}$ ou e^{-1} par leurs développements déduits de celui de e^x, puis qu'on multiplie les deux membres par $1.2.3\ldots n$, on trouvera aisément

$$\frac{a}{n+1}\left(1 + \frac{1}{n+2} + \ldots\right) \pm \frac{b}{n+1}\left(1 - \frac{1}{n+2} + \ldots\right) = \mu,$$

μ étant un entier. On peut toujours faire en sorte que le facteur

$$\pm \frac{b}{n+1}$$

soit positif; il suffira de supposer n pair si b est < 0 et n impair si b est > 0; en prenant de plus n très grand, l'équation que nous venons d'écrire conduira dès lors à une absurdité; car son premier membre étant essentiellement positif et très petit, sera compris entre 0 et 1, et ne pourra pas être égal à un entier μ. Donc, etc.

Liouville's paper

which is *approximately* $\frac{b}{n}$, so that for large n it certainly cannot be integral: It is larger than $\frac{b}{n+1}$ and smaller than $\frac{b}{n}$, as one can see from a comparison with a geometric series:

$$\begin{aligned}
\frac{1}{n+1} \;&<\; \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} + \frac{1}{(n+1)(n+2)(n+3)} + \ldots \\
&<\; \frac{1}{n+1} + \frac{1}{(n+1)^2} + \frac{1}{(n+1)^3} + \ldots \;=\; \frac{1}{n}.
\end{aligned}$$

Now one might be led to think that this simple multiply–by–$n!$ trick is not even sufficient to show that e^2 is irrational. This is a stronger statement: $\sqrt{2}$ is an example of a number which is irrational, but whose square is not.

From John Cosgrave we have learned that with two nice ideas/observations (let's call them "tricks") one can get two steps further nevertheless: Each of the tricks is sufficient to show that e^2 is irrational, the combination of both of them even yields the same for e^4. The first trick may be found in a one page paper by J. Liouville from 1840 — and the second one in a two page "addendum" which Liouville published on the next two journal pages.

Why is e^2 irrational? What can we derive from $e^2 = \frac{a}{b}$? According to Liouville we should write this as

$$be \;=\; ae^{-1},$$

substitute the series

$$e \;=\; 1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \ldots$$

and

$$e^{-1} \;=\; 1 - \frac{1}{1} + \frac{1}{2} - \frac{1}{6} + \frac{1}{24} - \frac{1}{120} \pm \ldots,$$

and then multiply by $n!$, for a sufficiently large even n. Then we see that $n!be$ is nearly integral:

$$n!b\Big(1 + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \ldots + \frac{1}{n!}\Big)$$

is an integer, and the rest

$$n!b\Big(\frac{1}{(n+1)!} + \frac{1}{(n+2)!} + \ldots\Big)$$

is approximately $\frac{b}{n}$: It is larger than $\frac{b}{n+1}$ but smaller than $\frac{b}{n}$, as we have seen above.

At the same time $n!ae^{-1}$ is nearly integral as well: Again we get a large integral part, and then a rest

$$(-1)^{n+1} n!a\Big(\frac{1}{(n+1)!} - \frac{1}{(n+2)!} + \frac{1}{(n+3)!} \mp \ldots\Big),$$

and this is approximately $(-1)^{n+1}\frac{a}{n}$. More precisely: for even n the rest is larger than $-\frac{a}{n}$, but smaller than

$$-a\Big(\frac{1}{n+1}-\frac{1}{(n+1)^2}-\frac{1}{(n+1)^3}-\ldots\Big) \;=\; -\frac{a}{n+1}\Big(1-\frac{1}{n}\Big) \;<\; 0.$$

But this cannot be true, since for large even n it would imply that $n!ae^{-1}$ is just a bit smaller than an integer, while $n!be$ is a bit larger than an integer, so $n!ae^{-1}=n!be$ cannot hold. $\square$

In order to show that e^4 is irrational, we now courageously assume that $e^4=\frac{a}{b}$ were rational, and write this as

$$be^2 \;=\; ae^{-2}.$$

We could now try to multiply this by $n!$ for some large n, and collect the non-integral summands, but this leads to nothing useful: The sum of the remaining terms on the left-hand side will be approximately $b\frac{2^{n+1}}{n}$, on the right side $(-1)^{n+1}a\frac{2^{n+1}}{n}$, and both will be very large if n gets large.

So one has to examine the situation a bit more carefully, and make two little adjustments to the strategy: First we will not take an *arbitrary* large n, but a large power of two, $n=2^m$; and secondly we will not multiply by $n!$, but by $\frac{n!}{2^{n-1}}$. Then we need a little lemma, a special case of Legendre's theorem (see page 8): For any $n\geq 1$ the integer $n!$ contains the prime factor 2 at most $n-1$ times — with equality if (and only if) n is a power of two, $n=2^m$.

This lemma is not hard to show: $\lfloor\frac{n}{2}\rfloor$ of the factors of $n!$ are even, $\lfloor\frac{n}{4}\rfloor$ of them are divisible by 4, and so on. So if 2^k is the largest power of two which satisfies $2^k\leq n$, then $n!$ contains the prime factor 2 exactly

$$\Big\lfloor\frac{n}{2}\Big\rfloor+\Big\lfloor\frac{n}{4}\Big\rfloor+\ldots+\Big\lfloor\frac{n}{2^k}\Big\rfloor \;\leq\; \frac{n}{2}+\frac{n}{4}+\ldots+\frac{n}{2^k} \;=\; n\Big(1-\frac{1}{2^k}\Big) \;\leq\; n-1$$

times, with equality in both inequalities exactly if $n=2^k$.

Let's get back to $be^2=ae^{-2}$. We are looking at

$$b\frac{n!}{2^{n-1}}e^2 \;=\; a\frac{n!}{2^{n-1}}e^{-2} \tag{1}$$

and substitute the series

$$e^2 \;=\; 1+\frac{2}{1}+\frac{4}{2}+\frac{8}{6}+\;\ldots\;+\frac{2^r}{r!}+\ldots$$

and

$$e^{-2} \;=\; 1-\frac{2}{1}+\frac{4}{2}-\frac{8}{6}\pm\;\ldots\;+(-1)^r\frac{2^r}{r!}+\ldots$$

For $r\leq n$ we get integral summands on both sides, namely

$$b\frac{n!}{2^{n-1}}\frac{2^r}{r!} \quad\text{resp.}\quad (-1)^r a\frac{n!}{2^{n-1}}\frac{2^r}{r!},$$

where for $r > 0$ the denominator $r!$ contains the prime factor 2 *at most* $r-1$ times, while $n!$ contains it *exactly* $n-1$ times. (So for $r > 0$ the summands are even.)

And since n is even (we assume that $n = 2^m$), the series that we get for $r \geq n+1$ are

$$2b\Big(\frac{2}{n+1} + \frac{4}{(n+1)(n+2)} + \frac{8}{(n+1)(n+2)(n+3)} + \dots\Big)$$

resp.

$$2a\Big(-\frac{2}{n+1} + \frac{4}{(n+1)(n+2)} - \frac{8}{(n+1)(n+2)(n+3)} \pm \dots\Big).$$

These series will for large n be roughly $\frac{4b}{n}$ resp. $-\frac{4a}{n}$, as one sees again by comparison with geometric series. For large $n = 2^m$ this means that the left-hand side of (1) is *a bit* larger than an integer, while the right-hand side is *a bit* smaller — contradiction! □

So we know that e^4 is irrational; to show that e^3, e^5 etc. are irrational as well, we need heavier machinery (that is, a bit of calculus), and a new idea — which essentially goes back to Charles Hermite, and for which the key is hidden in the following simple lemma.

Lemma. *For some fixed $n \geq 1$, let*

$$f(x) \;=\; \frac{x^n(1-x)^n}{n!}.$$

(i) *The function $f(x)$ is a polynomial of the form $f(x) = \dfrac{1}{n!}\displaystyle\sum_{i=n}^{2n} c_i x^i$, where the coefficients c_i are integers.*

(ii) *For $0 < x < 1$ we have $0 < f(x) < \frac{1}{n!}$.*

(iii) *The derivatives $f^{(k)}(0)$ and $f^{(k)}(1)$ are integers for all $k \geq 0$.*

■ **Proof.** Parts (i) and (ii) are clear.

For (iii) note that by (i) the k-th derivative $f^{(k)}$ vanishes at $x = 0$ unless $n \leq k \leq 2n$, and in this range $f^{(k)}(0) = \frac{k!}{n!}c_k$ is an integer. From $f(x) = f(1-x)$ we get $f^{(k)}(x) = (-1)^k f^{(k)}(1-x)$ for all x, and hence $f^{(k)}(1) = (-1)^k f^{(k)}(0)$, which is an integer. □

Theorem 1. *e^r is irrational for every $r \in \mathbb{Q}\backslash\{0\}$.*

■ **Proof.** It suffices to show that e^s cannot be rational for a positive integer s (if $e^{\frac{s}{t}}$ were rational, then $\left(e^{\frac{s}{t}}\right)^t = e^s$ would be rational, too). Assume that $e^s = \frac{a}{b}$ for integers $a, b > 0$, and let n be so large that $n! > as^{2n+1}$. Put

The estimate $n! > e(\frac{n}{e})^n$ yields an explicit n that is "large enough."

$$F(x) \;:=\; s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \dots + f^{(2n)}(x),$$

where $f(x)$ is the function of the lemma.

$F(x)$ may also be written as an infinite sum

$$F(x) \;=\; s^{2n}f(x) - s^{2n-1}f'(x) + s^{2n-2}f''(x) \mp \ldots,$$

since the higher derivatives $f^{(k)}(x)$, for $k > 2n$, vanish. From this we see that the polynomial $F(x)$ satisfies the identity

$$F'(x) \;=\; -s\,F(x) + s^{2n+1}f(x).$$

Thus differentiation yields

$$\frac{d}{dx}\left[e^{sx}F(x)\right] \;=\; se^{sx}F(x) + e^{sx}F'(x) \;=\; s^{2n+1}e^{sx}f(x)$$

and hence

$$N \;:=\; b\int_0^1 s^{2n+1}e^{sx}f(x)dx \;=\; b\left[e^{sx}F(x)\right]_0^1 \;=\; aF(1) - bF(0).$$

This is an integer, since part (iii) of the lemma implies that $F(0)$ and $F(1)$ are integers. However, part (ii) of the lemma yields estimates for the size of N from below and from above,

$$0 \;<\; N \;=\; b\int_0^1 s^{2n+1}e^{sx}f(x)dx \;<\; bs^{2n+1}e^s\frac{1}{n!} \;=\; \frac{as^{2n+1}}{n!} \;<\; 1,$$

which shows that N cannot be an integer: contradiction. □

Now that this trick was so successful, we use it once more.

Theorem 2. *π^2 is irrational.*

■ **Proof.** Assume that $\pi^2 = \frac{a}{b}$ for integers $a, b > 0$. We now use the polynomial

$$F(x) \;:=\; b^n\Big(\pi^{2n}f(x) - \pi^{2n-2}f^{(2)}(x) + \pi^{2n-4}f^{(4)}(x) \mp \ldots\Big),$$

which satisfies $F''(x) = -\pi^2F(x) + b^n\pi^{2n+2}f(x)$.

From part (iii) of the lemma we get that $F(0)$ and $F(1)$ are integers. Elementary differentiation rules yield

$$\begin{aligned}\frac{d}{dx}\big[F'(x)\sin\pi x - \pi F(x)\cos\pi x\big] \;&=\; \big(F''(x) + \pi^2F(x)\big)\sin\pi x\\ &=\; b^n\pi^{2n+2}f(x)\sin\pi x\\ &=\; \pi^2a^nf(x)\sin\pi x,\end{aligned}$$

and thus we obtain

$$\begin{aligned}N := \pi\int_0^1 a^nf(x)\sin\pi x\,dx \;&=\; \Big[\frac{1}{\pi}F'(x)\sin\pi x - F(x)\cos\pi x\Big]_0^1\\ &=\; F(0) + F(1),\end{aligned}$$

which is an integer. Furthermore N is positive since it is defined as the

π is not rational, but it does have "good approximations" by rationals — some of these were known since antiquity:

$$\begin{aligned}\tfrac{22}{7} &= 3.142857142857...\\ \tfrac{355}{113} &= 3.141592920353...\\ \tfrac{104348}{33215} &= 3.141592653921...\\ \pi &= 3.141592653589...\end{aligned}$$

integral of a function that is positive (except on the boundary). However, if we choose n so large that $\frac{\pi a^n}{n!} < 1$, then from part (ii) of the lemma we obtain

$$0 \;<\; N \;=\; \pi \int_0^1 a^n f(x) \sin \pi x \, dx \;<\; \frac{\pi a^n}{n!} \;<\; 1,$$

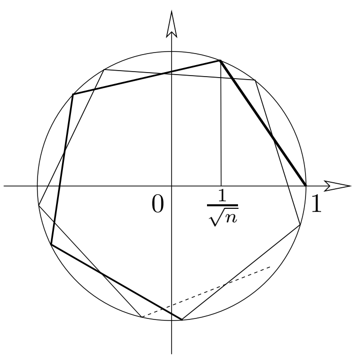
a contradiction. $\square$

Here comes our final irrationality result.

Theorem 3. *For every odd integer $n \geq 3$, the number*

$$A(n) \;\; := \;\; \frac{1}{\pi} \arccos\left(\frac{1}{\sqrt{n}}\right)$$

is irrational.



We will need this result for Hilbert's third problem (see Chapter 9) in the cases $n = 3$ and $n = 9$. For $n = 2$ and $n = 4$ we have $A(2) = \frac{1}{4}$ and $A(4) = \frac{1}{3}$, so the restriction to odd integers is essential. These values are easily derived by appealing to the diagram in the margin, in which the statement "$\frac{1}{\pi} \arccos\left(\frac{1}{\sqrt{n}}\right)$ is irrational" is equivalent to saying that the polygonal arc constructed from $\frac{1}{\sqrt{n}}$, all of whose chords have the same length, never closes into itself.

We leave it as an exercise for the reader to show that $A(n)$ is rational *only* for $n \in \{1, 2, 4\}$. For that, distinguish the cases when $n = 2^r$, and when n is not a power of 2.

■ **Proof.** We use the addition theorem

$$\cos \alpha + \cos \beta = 2 \cos \tfrac{\alpha+\beta}{2} \cos \tfrac{\alpha-\beta}{2}$$

from elementary trigonometry, which for $\alpha = (k+1)\varphi$ and $\beta = (k-1)\varphi$ yields

$$\cos{(k+1)\varphi} \;\; = \;\; 2 \cos\varphi \, \cos k\varphi \; - \; \cos{(k-1)\varphi}. \qquad (2)$$

For the angle $\varphi_n = \arccos\left(\frac{1}{\sqrt{n}}\right)$, which is defined by $\cos\varphi_n = \frac{1}{\sqrt{n}}$ and $0 \leq \varphi_n \leq \pi$, this yields representations of the form

$$\cos k\varphi_n \;\; = \;\; \frac{A_k}{\sqrt{n}^k},$$

where A_k is an integer that is not divisible by n, for all $k \geq 0$. In fact, we have such a representation for $k = 0, 1$ with $A_0 = A_1 = 1$, and by induction on k using (2) we get for $k \geq 1$

$$\cos{(k+1)\varphi_n} \;\; = \;\; 2 \frac{1}{\sqrt{n}} \frac{A_k}{\sqrt{n}^k} - \frac{A_{k-1}}{\sqrt{n}^{k-1}} \;\; = \;\; \frac{2A_k - nA_{k-1}}{\sqrt{n}^{k+1}}.$$

Thus we obtain $A_{k+1} = 2A_k - nA_{k-1}$. If $n \geq 3$ is odd, and A_k is not divisible by n, then we find that A_{k+1} cannot be divisible by n, either.

Now assume that

$$A(n) \;=\; \frac{1}{\pi}\varphi_n \;=\; \frac{k}{\ell}$$

is rational (with integers $k, \ell > 0$). Then $\ell\varphi_n = k\pi$ yields

$$\pm 1 \;=\; \cos k\pi \;=\; \frac{A_\ell}{\sqrt{n}^{\ell}}.$$

Thus $\sqrt{n}^{\ell} = \pm A_\ell$ is an integer, with $\ell \geq 2$, and hence $n \,|\, \sqrt{n}^{\ell}$. With $\sqrt{n}^{\ell} \,|\, A_\ell$ we find that n divides A_ℓ, a contradiction. $\square$

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Three times $\pi^2/6$ — Chapter 8

We know that the infinite series $\sum_{n\geq 1} \frac{1}{n}$ does not converge. Indeed, in Chapter 1 we have seen that even the series $\sum_{p\in\mathbb{P}} \frac{1}{p}$ diverges.

However, the sum of the reciprocals of the squares converges (although very slowly, as we will also see), and it produces an interesting value.

Euler's series.

$$\sum_{n\geq 1} \frac{1}{n^2} \;=\; \frac{\pi^2}{6}.$$

This is a classical, famous and important result by Leonhard Euler from 1734. One of its key interpretations is that it yields the first non-trivial value $\zeta(2)$ of Riemann's zeta function (see the appendix on page 49). This value is irrational, as we have seen in Chapter 7.

But not only the result has a prominent place in mathematics history, there are also a number of extremely elegant and clever proofs that have their history: For some of these the joy of discovery and rediscovery has been shared by many. In this chapter, we present three such proofs.

$$\begin{array}{lcl}
1 & = & 1.000000 \\
1+\frac{1}{4} & = & 1.250000 \\
1+\frac{1}{4}+\frac{1}{9} & = & 1.361111 \\
1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16} & = & 1.423611 \\
1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\frac{1}{25} & = & 1.463611 \\
1+\frac{1}{4}+\frac{1}{9}+\frac{1}{16}+\frac{1}{25}+\frac{1}{36} & = & 1.491388 \\
\pi^2/6 & = & 1.644934.
\end{array}$$

■ **Proof.** The first proof appears as an exercise in William J. LeVeque's number theory textbook from 1956. But he says: "I haven't the slightest idea where that problem came from, but I'm pretty certain that it wasn't original with me."

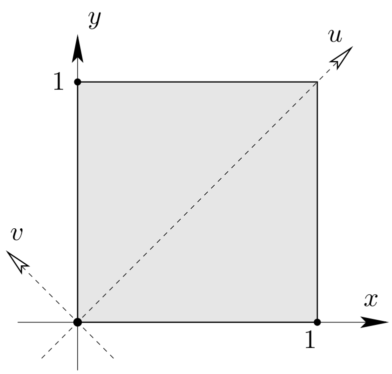
The proof consists in two different evaluations of the double integral

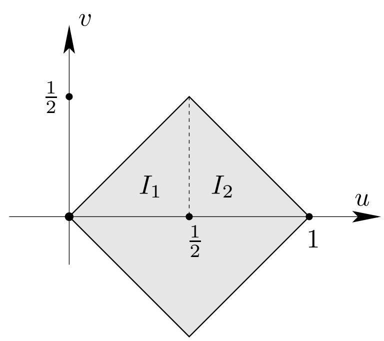
$$I \;:=\; \int_0^1\!\!\int_0^1 \frac{1}{1-xy}\, dx\, dy.$$

For the first one, we expand $\frac{1}{1-xy}$ as a geometric series, decompose the summands as products, and integrate effortlessly:

$$\begin{aligned}
I \;&=\; \int_0^1\!\!\int_0^1 \sum_{n\geq 0} (xy)^n\, dx\, dy \;=\; \sum_{n\geq 0} \int_0^1\!\!\int_0^1 x^n y^n\, dx\, dy \\
&=\; \sum_{n\geq 0} \Big(\int_0^1 x^n dx\Big)\Big(\int_0^1 y^n dy\Big) \;=\; \sum_{n\geq 0} \frac{1}{n+1}\,\frac{1}{n+1} \\
&=\; \sum_{n\geq 0} \frac{1}{(n+1)^2} \;=\; \sum_{n\geq 1} \frac{1}{n^2} \;=\; \zeta(2).
\end{aligned}$$

This evaluation also shows that the double integral (over a positive function with a pole at $x = y = 1$) is finite. Note that the computation is also easy and straightforward if we read it backwards — thus the evaluation of $\zeta(2)$ leads one to the double integral I.





The second way to evaluate I comes from a change of coordinates: in the new coordinates given by $u := \frac{y+x}{2}$ and $v := \frac{y-x}{2}$ the domain of integration is a square of side length $\frac{1}{2}\sqrt{2}$, which we get from the old domain by first rotating it by 45° and then shrinking it by a factor of $\sqrt{2}$. Substitution of $x = u - v$ and $y = u + v$ yields

$$\frac{1}{1-xy} \;=\; \frac{1}{1-u^2+v^2}.$$

To transform the integral, we have to replace $dx\,dy$ by $2\,du\,dv$, to compensate for the fact that our coordinate transformation reduces areas by a constant factor of 2 (which is the Jacobi determinant of the transformation; see the box on the next page). The new domain of integration, and the function to be integrated, are symmetric with respect to the u-axis, so we just need to compute two times (another factor of 2 arises here!) the integral over the upper half domain, which we split into two parts in the most natural way:

$$I \;=\; 4\int_0^{1/2}\Big(\int_0^u \frac{dv}{1-u^2+v^2}\Big)du \;+\; 4\int_{1/2}^1\Big(\int_0^{1-u}\frac{dv}{1-u^2+v^2}\Big)du.$$

Using $\displaystyle\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\frac{x}{a} + C$, this becomes

$$\begin{aligned} I \;=\;& \quad 4\int_0^{1/2}\frac{1}{\sqrt{1-u^2}}\arctan\Big(\frac{u}{\sqrt{1-u^2}}\Big)\,du \\ &+\; 4\int_{1/2}^1\frac{1}{\sqrt{1-u^2}}\arctan\Big(\frac{1-u}{\sqrt{1-u^2}}\Big)\,du. \end{aligned}$$

These integrals can be simplified and finally evaluated by substituting $u = \sin\theta$ resp. $u = \cos\theta$. But we proceed more directly, by computing that the derivative of $g(u) := \arctan\big(\frac{u}{\sqrt{1-u^2}}\big)$ is $g'(u) = \frac{1}{\sqrt{1-u^2}}$, while the derivative of $h(u) := \arctan\big(\frac{1-u}{\sqrt{1-u^2}}\big) = \arctan\Big(\sqrt{\frac{1-u}{1+u}}\Big)$ is $h'(u) = -\frac{1}{2}\frac{1}{\sqrt{1-u^2}}$. So we may use $\int_a^b f'(x)f(x)dx = \big[\frac{1}{2}f(x)^2\big]_a^b = \frac{1}{2}f(b)^2 - \frac{1}{2}f(a)^2$ and get

$$\begin{aligned} I &= 4\int_0^{1/2} g'(u)g(u)\,du \;+\; 4\int_{1/2}^1 -2h'(u)h(u)\,du \\ &= 2\Big[g(u)^2\Big]_0^{1/2} \;-\; 4\Big[h(u)^2\Big]_{1/2}^1 \\ &= 2g(\tfrac{1}{2})^2 \;-\; 2g(0)^2 \;-\; 4h(1)^2 \;+\; 4h(\tfrac{1}{2})^2 \\ &= 2\big(\tfrac{\pi}{6}\big)^2 - 0 - 0 + 4\big(\tfrac{\pi}{6}\big)^2 \;=\; \tfrac{\pi^2}{6}. \end{aligned}$$

$\square$

This proof extracted the value of Euler's series from an integral via a rather simple coordinate transformation. An ingenious proof of this type — with an entirely non-trivial coordinate transformation — was later discovered by Beukers, Calabi and Kolk. The point of departure for that proof is to split the sum $\sum_{n\geq 1}\frac{1}{n^2}$ into the even terms and the odd terms. Clearly the even terms $\frac{1}{2^2}+\frac{1}{4^2}+\frac{1}{6^2}+\ldots=\sum_{k\geq 1}\frac{1}{(2k)^2}$ sum to $\frac{1}{4}\zeta(2)$, so the odd terms $\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\ldots=\sum_{k\geq 0}\frac{1}{(2k+1)^2}$ make up three quarters of the total sum $\zeta(2)$. Thus Euler's series is equivalent to

$$\sum_{k\geq 0}\frac{1}{(2k+1)^2} \;=\; \frac{\pi^2}{8}.$$

■ **Proof.** As above, we may express this as a double integral, namely

$$J \;=\; \int_0^1\!\!\int_0^1 \frac{1}{1-x^2y^2}\,dx\,dy \;=\; \sum_{k\geq 0}\frac{1}{(2k+1)^2}.$$

So we have to compute this integral J. And for this Beukers, Calabi and Kolk proposed the new coordinates

$$u \;:=\; \arccos\sqrt{\frac{1-x^2}{1-x^2y^2}} \qquad v \;:=\; \arccos\sqrt{\frac{1-y^2}{1-x^2y^2}}.$$

To compute the double integral, we may ignore the boundary of the domain, and consider x, y in the range $0<x<1$ and $0<y<1$. Then u, v will lie in the triangle $u>0$, $v>0$, $u+v<\pi/2$. The coordinate transformation can be inverted explicitly, which leads one to the substitution

$$x \;=\; \frac{\sin u}{\cos v} \qquad \text{and} \qquad y \;=\; \frac{\sin v}{\cos u}.$$

It is easy to check that these formulas define a bijective coordinate transformation between the interior of the unit square $S=\{(x,y):0\leq x,y\leq 1\}$ and the interior of the triangle $T=\{(u,v):u,v\geq 0,\ u+v\leq\pi/2\}$.

Now we have to compute the Jacobi determinant of the coordinate transformation, and magically it turns out to be

$$\det\begin{pmatrix}\frac{\cos u}{\cos v} & \frac{\sin u\sin v}{\cos^2 v}\\ \frac{\sin u\sin v}{\cos^2 u} & \frac{\cos v}{\cos u}\end{pmatrix} \;=\; 1-\frac{\sin^2 u\sin^2 v}{\cos^2 u\cos^2 v} \;=\; 1-x^2y^2.$$

But this means that the integral that we want to compute is transformed into

$$J \;=\; \int_0^{\pi/2}\int_0^{\pi/2-u} 1\,du\,dv,$$

which is just the area $\frac{1}{2}(\frac{\pi}{2})^2=\frac{\pi^2}{8}$ of the triangle T. □

The Substitution Formula

To compute a double integral

$$I \;=\; \int_S f(x,y)\,dx\,dy.$$

we may perform a substitution of variables

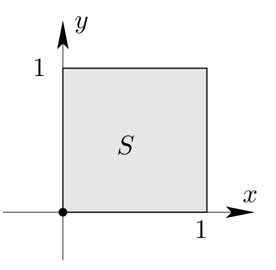
$$x=x(u,v) \quad y=y(u,v),$$

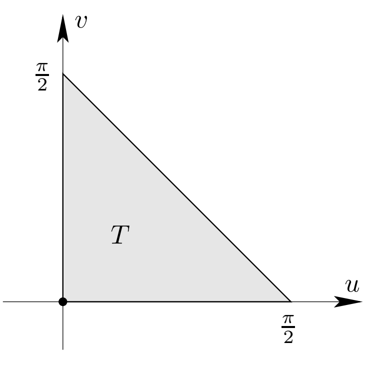
if the correspondence of $(u,v)\in T$ to $(x,y)\in S$ is bijective and continuously differentiable. Then I equals

$$\int_T f(x(u,v),y(u,v))\left|\frac{d(x,y)}{d(u,v)}\right|du\,dv,$$

where $\frac{d(x,y)}{d(u,v)}$ is the Jacobi determinant:

$$\frac{d(x,y)}{d(u,v)} \;=\; \det\begin{pmatrix}\frac{dx}{du} & \frac{dx}{dv}\\ \frac{dy}{du} & \frac{dy}{dv}\end{pmatrix}.$$





Beautiful — even more so, as the same method of proof extends to the computation of $\zeta(2k)$ in terms of a $2k$-dimensional integral, for all $k \geq 1$. We refer to the original paper of Beuker, Calabi and Kolk [2], and to Chapter 23, where we'll achieve this on a different path, using the Herglotz trick and Euler's original approach.

After these two proofs via coordinate transformation we can't resist the temptation to present another, entirely different and completely elementary proof for $\sum_{n\geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}$. It appears in a sequence of exercises in the problem book by the twin brothers Akiva and Isaak Yaglom, whose Russian original edition appeared in 1954. Versions of this beautiful proof were rediscovered and presented by F. Holme (1970), I. Papadimitriou (1973), and by Ransford (1982) who attributed it to John Scholes.

■ **Proof.** The first step is to establish a remarkable relation between values of the (squared) cotangent function. Namely, for all $m \geq 1$ one has

$$\cot^2\left(\tfrac{\pi}{2m+1}\right) + \cot^2\left(\tfrac{2\pi}{2m+1}\right) + \ldots + \cot^2\left(\tfrac{m\pi}{2m+1}\right) \;=\; \tfrac{2m(2m-1)}{6}. \qquad (1)$$

For $m = 1, 2, 3$ this yields
$\cot^2 \frac{\pi}{3} = \frac{1}{3}$
$\cot^2 \frac{\pi}{5} + \cot^2 \frac{2\pi}{5} = 2$
$\cot^2 \frac{\pi}{7} + \cot^2 \frac{2\pi}{7} + \cot^2 \frac{3\pi}{7} = 5$

To establish this, we start with the relation $e^{ix} = \cos x + i \sin x$. Taking the n-th power $e^{inx} = (e^{ix})^n$, we get

$$\cos nx + i \sin nx \;=\; (\cos x + i \sin x)^n.$$

The imaginary part of this is

$$\sin nx \;=\; \binom{n}{1} \sin x \cos^{n-1} x - \binom{n}{3} \sin^3 x \cos^{n-3} x \pm \ldots \qquad (2)$$

Now we let $n = 2m + 1$, while for x we will consider the m different values $x = \frac{r\pi}{2m+1}$, for $r = 1, 2, \ldots, m$. For each of these values we have $nx = r\pi$, and thus $\sin nx = 0$, while $0 < x < \frac{\pi}{2}$ implies that for $\sin x$ we get m distinct positive values.

In particular, we can divide (2) by $\sin^n x$, which yields

$$0 \;=\; \binom{n}{1} \cot^{n-1} x - \binom{n}{3} \cot^{n-3} x \pm \ldots,$$

that is,

$$0 \;=\; \binom{2m+1}{1} \cot^{2m} x - \binom{2m+1}{3} \cot^{2m-2} x \pm \ldots$$

for each of the m distinct values of x. Thus for the polynomial of degree m

$$p(t) \;:=\; \binom{2m+1}{1} t^m - \binom{2m+1}{3} t^{m-1} \pm \ldots + (-1)^m \binom{2m+1}{2m+1}$$

we know m distinct roots

$$a_r \;=\; \cot^2\left(\tfrac{r\pi}{2m+1}\right) \quad \text{for} \quad r = 1, 2, \ldots, m.$$

Hence the polynomial coincides with

$$p(t) \;=\; \binom{2m+1}{1} \left(t - \cot^2\left(\tfrac{\pi}{2m+1}\right)\right) \cdots \left(t - \cot^2\left(\tfrac{m\pi}{2m+1}\right)\right).$$

Comparison of the coefficients of t^{m-1} in $p(t)$ now yields that the sum of the roots is

$$a_1 + \ldots + a_r \;=\; \frac{\binom{2m+1}{3}}{\binom{2m+1}{1}} \;=\; \frac{2m(2m-1)}{6},$$

which proves (1).

We also need a second identity, of the same type,

$$\csc^2\left(\tfrac{\pi}{2m+1}\right) + \csc^2\left(\tfrac{2\pi}{2m+1}\right) + \ldots + \csc^2\left(\tfrac{m\pi}{2m+1}\right) = \frac{2m(2m+2)}{6}, \qquad (3)$$

for the cosecant function $\csc x = \frac{1}{\sin x}$. But

$$\csc^2 x = \frac{1}{\sin^2 x} = \frac{\cos^2 x + \sin^2 x}{\sin^2 x} = \cot^2 x + 1,$$

so we can derive (3) from (1) by adding m to both sides of the equation.

Now the stage is set, and everything falls into place. We use that in the range $0 < y < \frac{\pi}{2}$ we have

$$0 \;<\; \sin y \;<\; y \;<\; \tan y,$$

and thus

$$0 \;<\; \cot y \;<\; \tfrac{1}{y} \;<\; \csc y,$$

which implies

$$\cot^2 y \;<\; \tfrac{1}{y^2} \;<\; \csc^2 y.$$

Now we take this double inequality, apply it to each of the m distinct values of x, and add the results. Using (1) for the left-hand side, and (3) for the right-hand side, we obtain

$$\frac{2m(2m-1)}{6} \;<\; \left(\tfrac{2m+1}{\pi}\right)^2 + \left(\tfrac{2m+1}{2\pi}\right)^2 + \ldots + \left(\tfrac{2m+1}{m\pi}\right)^2 \;<\; \frac{2m(2m+2)}{6},$$

that is,

$$\frac{\pi^2}{6}\frac{2m}{2m+1}\frac{2m-1}{2m+1} \;<\; \frac{1}{1^2} + \frac{1}{2^2} + \ldots + \frac{1}{m^2} \;<\; \frac{\pi^2}{6}\frac{2m}{2m+1}\frac{2m+2}{2m+1}.$$

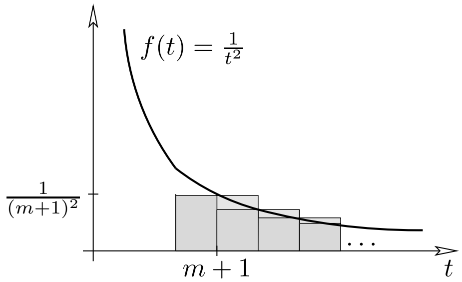
Both the left-hand and the right-hand side converge to $\frac{\pi^2}{6}$ for $m \longrightarrow \infty$: end of proof. $\square$

So how fast does $\sum \frac{1}{n^2}$ converge to $\pi^2/6$? For this we have to estimate the difference

$$\frac{\pi^2}{6} - \sum_{n=1}^{m} \frac{1}{n^2} \;=\; \sum_{n=m+1}^{\infty} \frac{1}{n^2}.$$

Comparison of coefficients:
If $p(t) = c(t-a_1)\cdots(t-a_m)$, then the coefficient of t^{m-1} is $-c(a_1 + \ldots + a_m)$.

$0 < a < b < c$ implies $0 < \frac{1}{c} < \frac{1}{b} < \frac{1}{a}$



This is very easy with the technique of "comparing with an integral" that we have reviewed already in the appendix to Chapter 2 (page 10). It yields

$$\sum_{n=m+1}^{\infty} \frac{1}{n^2} \;<\; \int_m^\infty \frac{1}{t^2}dt \;=\; \frac{1}{m}$$

for an upper bound and

$$\sum_{n=m+1}^{\infty} \frac{1}{n^2} \;>\; \int_{m+1}^\infty \frac{1}{t^2}dt \;=\; \frac{1}{m+1}$$

for a lower bound on the "remaining summands" — or even

$$\sum_{n=m+1}^{\infty} \frac{1}{n^2} \;>\; \int_{m+\frac{1}{2}}^\infty \frac{1}{t^2}dt \;=\; \frac{1}{m+\frac{1}{2}}$$

if you are willing to do a slightly more careful estimate, using that the function $f(t) = \frac{1}{t^2}$ is convex.

This means that our series does not converge too well; if we sum the first one thousand summands, then we expect an error in the third digit after the decimal point, while for the sum of the first one million summands, $m = 1000000$, we expect to get an error in the sixth decimal digit, and we do. However, then comes a big surprise: to an accuracy of 45 digits,

$$\begin{aligned} \pi^2/6 \;&=\; 1.644934066848226436472415166646025189218949901,\\ \sum_{n=1}^{10^6} \frac{1}{n^2} \;&=\; 1.644933066848726436305748499979391855885616544. \end{aligned}$$

So the sixth digit after the decimal point is wrong (too small by 1), but *the next six digits are right*! And then one digit is wrong (too large by 5), then again five are correct. This surprising discovery is quite recent, due to Roy D. North from Colorado Springs, 1988. (In 1982, Martin R. Powell, a school teacher from Amersham, Bucks, England, failed to notice the full effect due to the insufficient computing power available at the time.) It is too strange to be purely coincidental … A look at the error term, which again to 45 digits reads

$$\sum_{n=10^6+1}^{\infty} \frac{1}{n^2} \;=\; 0.000000999999500000166666666666633333333333357,$$

reveals that clearly there is a pattern. You might try to rewrite this last number as

$$+\,10^{-6} \;-\; \tfrac{1}{2}10^{-12} \;+\; \tfrac{1}{6}10^{-18} \;-\; \tfrac{1}{30}10^{-30} \;+\; \tfrac{1}{42}10^{-42} \;+\; \dots$$

where the coefficients $(1, -\frac{1}{2}, \frac{1}{6}, 0, -\frac{1}{30}, 0, \frac{1}{42})$ of 10^{-6i} form the beginning of the sequence of *Bernoulli numbers* that we'll meet again in Chapter 23. We refer our readers to the article by Borwein, Borwein & Dilcher [3] for more such surprising "coincidences" — and for proofs.

Appendix: The Riemann zeta function

The *Riemann zeta function* $\zeta(s)$ is defined for real $s > 1$ by

$$\zeta(s) \ := \ \sum_{n\geq 1} \frac{1}{n^s}.$$

Our estimates for H_n (see page 10) imply that the series for $\zeta(1)$ diverges, but for any real $s > 1$ it does converge. The zeta function has a canonical continuation to the entire complex plane (with one simple pole at $s = 1$), which can be constructed using power series expansions. The resulting complex function is of utmost importance for the theory of prime numbers. Let us mention four diverse connections:

(1) The remarkable identity

$$\zeta(s) \ = \ \prod_p \frac{1}{1-p^{-s}}$$

is due to Euler. It encodes the basic fact that every natural number has a unique (!) decomposition into prime factors; using this, Euler's identity is a simple consequence of the geometric series expansion

$$\frac{1}{1-p^{-s}} \ = \ 1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \frac{1}{p^{3s}} + \dots$$

(2) The following marvelous argument of Don Zagier computes $\zeta(4)$ from $\zeta(2)$. Consider the function

$$f(m,n) \ = \ \frac{2}{m^3 n} + \frac{1}{m^2 n^2} + \frac{2}{m n^3}$$

for integers $m, n \geq 1$. It is easily verified that for all m and n,

$$f(m,n) - f(m+n,n) - f(m,m+n) \ = \ \frac{2}{m^2 n^2}.$$

Let us sum this equation over all $m, n \geq 1$. If $i \neq j$, then (i,j) is either of the form $(m+n,n)$ or of the form $(m,m+n)$, for $m, n \geq 1$. Thus, in the sum on the left-hand side all terms $f(i,j)$ with $i \neq j$ cancel, and so

$$\sum_{n\geq 1} f(n,n) \ = \ \sum_{n\geq 1} \frac{5}{n^4} \ = \ 5\zeta(4)$$

remains. For the right-hand side one obtains

$$\sum_{m,n\geq 1} \frac{2}{m^2 n^2} \ = \ 2 \sum_{m\geq 1} \frac{1}{m^2} \cdot \sum_{n\geq 1} \frac{1}{n^2} \ = \ 2\zeta(2)^2,$$

and out comes the equality

$$5\zeta(4) \ = \ 2\zeta(2)^2.$$

With $\zeta(2) = \frac{\pi^2}{6}$ we thus get $\zeta(4) = \frac{\pi^4}{90}$.
Another derivation via Bernoulli numbers appears in Chapter 23.

(3) It has been known for a long time that $\zeta(s)$ is a rational multiple of π^s, and hence irrational, if s is an *even* integer $s \geq 2$; see Chapter 23. In contrast, the irrationality of $\zeta(3)$ was proved by Roger Apéry only in 1979. Despite considerable effort the picture is rather incomplete about $\zeta(s)$ for the other odd integers, $s = 2t+1 \geq 5$. Very recently, Keith Ball and Tanguy Rivoal proved that infinitely many of the values $\zeta(2t+1)$ are irrational. And indeed, although it is not known for any single odd value $s \geq 5$ that $\zeta(s)$ is irrational, Wadim Zudilin has proved that at least one of the four values $\zeta(5)$, $\zeta(7)$, $\zeta(9)$, and $\zeta(11)$ is irrational. We refer to the beautiful survey by Fischler.

(4) The location of the complex zeros of the zeta function is the subject of the "Riemann hypothesis": one of the most famous and important unresolved conjectures in all of mathematics. It claims that all the non-trivial zeros $s \in \mathbb{C}$ of the zeta function satisfy $\mathrm{Re}(s) = \frac{1}{2}$. (The zeta function vanishes at all the negative even integers, which are referred to as the "trivial zeros.")

Very recently, Jeff Lagarias showed that, surprisingly, the Riemann hypothesis is equivalent to the following elementary statement: For all $n \geq 1$,

$$\sum_{d \mid n} d \;\leq\; H_n \,+\, \exp(H_n)\log(H_n),$$

with equality only for $n = 1$, where H_n is again the n-th harmonic number.

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